

IKG Studio — Model Training Platform User Manual

Version 2.0

This manual documents the web UI of **IKG Studio Model Training Platform**, a browser-based tool for registering image datasets, training / converting / benchmarking computer-vision models, and monitoring background jobs. It does **not** cover deployment or infrastructure setup.

Screenshots are stored in `docs/manual-screenshots/`.

Scope of this build. Only **detection (DETECT)** and **oriented bounding box (OBB)** tasks are supported. SEGMENT, POSE and CLASSIFY appear in the data model but are reported as disabled by the platform and are rejected by the dataset scanner.

Table of Contents

- [1. Concepts & Object Model](#)
- [2. Sign In](#)
- [3. Roles & Permissions](#)
- [4. Navigation](#)
- [5. Home \(Dashboard\)](#)
- [6. Datasets](#)
 - [6.1 Source Datasets](#)
 - [6.2 Training Datasets](#)
- [7. Models](#)
- [8. Downloading a Trained Model](#)
- [9. Training](#)
- [10. Benchmarks](#)
- [11. Jobs](#)
- [12. Notifications](#)
- [13. Admin](#)
 - [13.1 Users](#)
 - [13.2 Dataset Types](#)
 - [13.3 Audit](#)
 - [13.4 System Settings](#)
 - [13.5 Workers](#)
 - [13.6 Backup](#)
- [14. Account & Security](#)
- [15. Status Reference](#)
- [16. Troubleshooting & Edge Cases](#)
- [17. End-to-End Workflow](#)

1. Concepts & Object Model

Five objects carry the whole platform. Everything else is a view over them.

Object	What it is	Created by
Dataset Type	A category (cards , dice , ...) that scopes everything else and carries three filesystem roots. Types form a tree ; a child inherits its parent's paths unless it sets its own.	Admin
Source Dataset	One raw image folder discovered under a type's <i>dataset path</i> , plus the result of scanning it (image/label pairs, class list, issues). Read-only — the platform never writes into it.	Registering + scanning a folder
Training Dataset	A curated YOLO dataset with <code>train / val / test</code> splits and a <code>data.yaml</code> . Either BUILT from one or more source datasets, or REGISTERED by pointing at a directory that is already laid out.	The New Training Dataset wizard
Model	A <code>.pt</code> checkpoint plus its metrics, hyperparameters and lineage. Either trained on the	A training job,

	platform or discovered by scanning a type's <i>model path</i> .	or Scan Model Roots
Job / Execution	A job is the unit of work you asked for (train this, build that). An execution is one attempt at it. A job can have several executions — a retry adds an attempt, it does not create a new job.	Every long-running action

The chain is: **Dataset Type → Source Datasets → Training Dataset → Training Job → Model → Benchmark Run.**

Where your data lives

	What it holds
PostgreSQL	All state: types, datasets, scans, jobs, executions, models, benchmark runs, audit, notifications, settings
Object storage (MinIO)	Artifacts: trained weights, training logs, charts, <code>results.csv</code> , benchmark outputs, OpenVINO <code>.zip</code> exports
Mounted filesystem	Three roots per dataset type — dataset path (source image folders, read-only), model path (<code>.pt</code> files, the "Model Root"), training dataset path (built / registered YOLO directories)

A trained checkpoint therefore exists in **two** places: as a downloadable artifact in object storage, and as a file under the type's Model Root on the server. [Section 8](#) explains how to get at both.

2. Sign In

Navigate to the platform URL. The login screen offers a username/password form and a passkey (WebAuthn) button.



IKG Studio

Model Training Platform

Username

Password

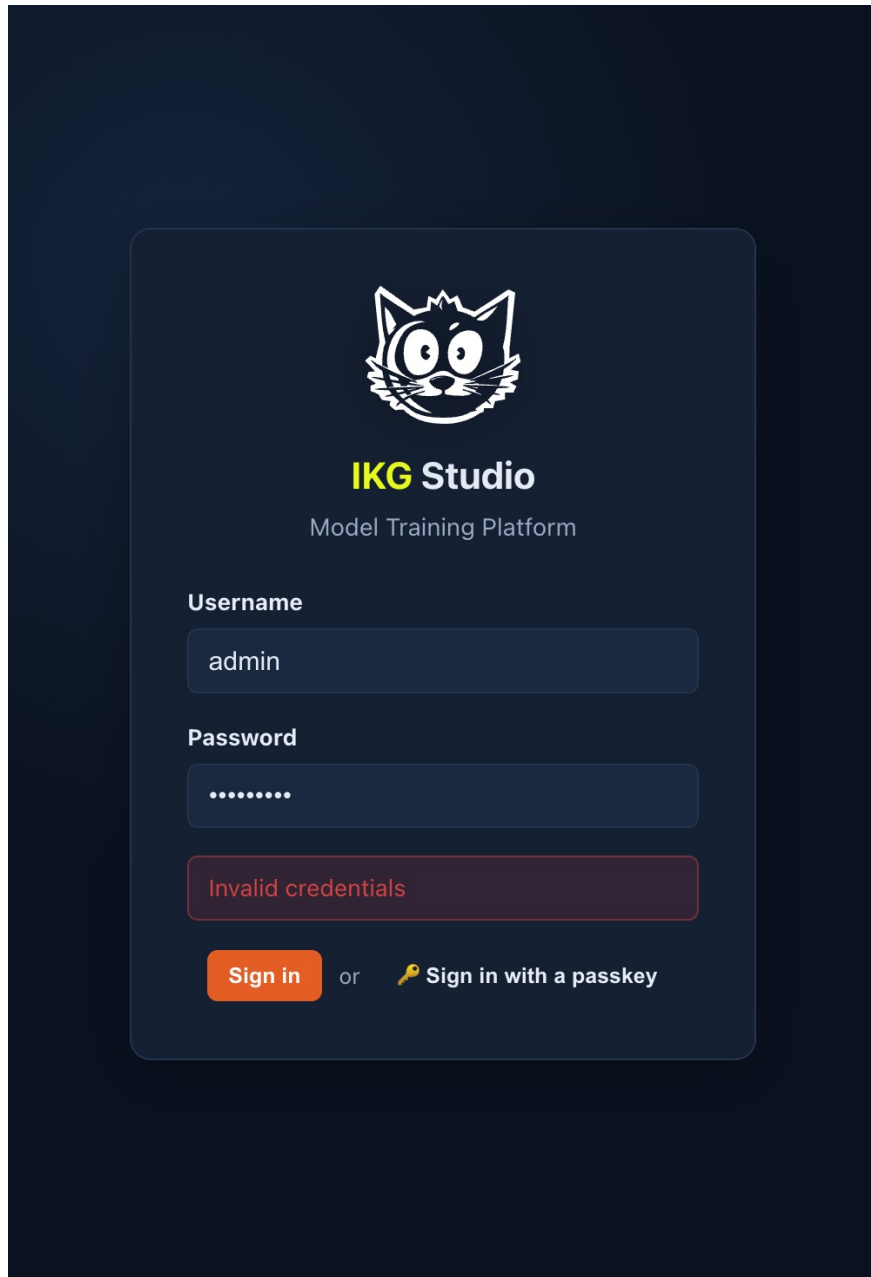
Sign in

or



Sign in with a passkey

Enter credentials and click **Sign in**. On failure an inline error is shown and the form stays editable:



Account logout. After `auth_failed_login_threshold` failed attempts (default 5) the account is locked for `auth_lockout_minutes` (default 15). A locked account shows status `LOCKED` on the Admin → Users page and an admin can clear it early with **Unlock**. See [System Settings](#).

Sessions expire on two independent clocks: `auth_session_idle_minutes` (default 480) and `auth_session_absolute_hours` (default 24).

3. Roles & Permissions

There are two roles.

Capability	ADMIN	USER
Source datasets — register, rescan, class override, archive	✓	✓
Training datasets — create, build, register, re-validate	✓	✓
Delete a training dataset	✓	✗
Models — view, Scan Model Roots, download artifacts	✓	✓
Delete a model	✓	✗

Create / delete an OpenVINO conversion	✓	✗
Training jobs — create, stop, retry	✓	✓
Benchmark runs — create, stop, retry	✓	✓
Jobs, Notifications, per-resource History	✓	✓
Workers list	✓	✓
Admin section (Users, Dataset Types, Audit, System Settings, Backup)	✓	✗
Browse the server filesystem (path pickers)	✓	✗

Earlier versions of this manual said a USER could do everything an admin could except open the Admin tab. That is **not** correct: deleting a model, deleting a training dataset, and creating or deleting an OpenVINO conversion are admin-only. The buttons are hidden for USERS and the API rejects the calls with HTTP 403.

Note also that **benchmark runs cannot be deleted by anyone** — there is no delete endpoint. A run can only be stopped, cancelled or retried.

The **Admin** nav entry is rendered only for ADMIN sessions. If a stale "admin" page survives a user switch on the same browser, the shell sends you back to Home. Browsing state is reset on every sign-in and sign-out, so nothing leaks between accounts.

 IKG Studio

Home

Datasets

Models

Training

Benchmarks

Jobs

Notifications

USER user

Sign Out

Dashboard

328
Source Datasets

2
Training Datasets

3
Models

4
Training Jobs

4
Benchmarks

System Health

2
Workers Online

0
Workers Offline

2
Workers Total

0
Active Executions

0
Pending Outbox

0
Dead-Letter Outbox

Active Jobs

No active jobs

Recent Models

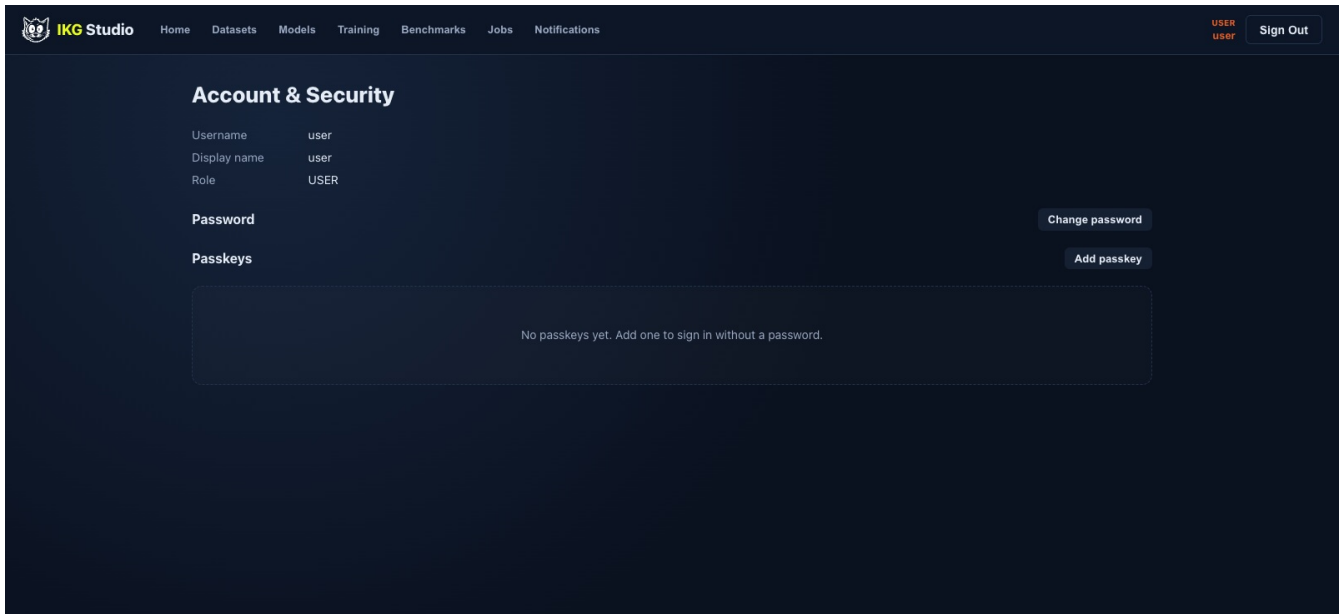
NAME	TASK	SOURCE	STATUS	CREATED
dice_I_X	DETECT	TRAINING	AVAILABLE	Aug 06, 2026, 08:11
cards 8/6/2026	DETECT	TRAINING	AVAILABLE	Aug 06, 2026, 12:22
cards 8/5/2026	DETECT	TRAINING	AVAILABLE	Aug 06, 2026, 10:53

Recent Benchmarks

NAME	STATUS	PROGRESS	CREATED
QA Compare Seed	COMPLETED	1/1	Aug 06, 2026, 01:28 PM
ts-verify	COMPLETED	1/1	Aug 06, 2026, 11:40 AM
test	COMPLETED	1/1	Aug 06, 2026, 11:19 AM
test1	COMPLETED	1/1	Aug 06, 2026, 10:43 AM

Recent Activity

ACTION	RESOURCE	RESULT	WHEN
AUTH_LOGIN_SUCCEEDED	USER	SUCCESS	Aug 07, 2026, 09:44 AM
AUTH_LOGIN_SUCCEEDED	USER	SUCCESS	Aug 06, 2026, 04:48 PM



4. Navigation

The top app bar contains the brand, the primary nav (**Home • Datasets • Models • Training • Benchmarks • Jobs • Notifications**, plus **Admin** for admins), your role badge, an account button and **Sign Out**.

There is no URL router. The page you are on is remembered in browser storage, not in the address bar. Query parameters such as `?modelId=...`, `?trainingJobId=...`, `?jobId=...` only select *which record* the currently open page shows — they cannot switch pages. A link with `?modelId=...` therefore only works if the recipient is already on the Models tab.

Back remembers where you came from. Jumping model → training job and pressing ← **Back** returns to the model, not to the training list. The same applies to training dataset → source dataset, and model → training dataset.

Live updates. The app holds a server-sent-event stream open and also polls (jobs, models and datasets every 5 s; the unread badge and dashboard every 20 s; an open log tail every 3 s). Job states, notification badges and conversion progress update on their own — you do not need to refresh.

5. Home (Dashboard)

 IKG Studio

[Home](#) [Datasets](#) [Models](#) [Training](#) [Benchmarks](#) [Jobs](#) [Notifications](#) [Admin](#)

[System Admin](#) [Sign Out](#)

Dashboard

328
Source Datasets

2
Training Datasets

3
Models

4
Training Jobs

4
Benchmarks

System Health

2
Workers Online

0
Workers Offline

2
Workers Total

0
Active Executions

0
Pending Outbox

0
Dead-Letter Outbox

Active Jobs

No active jobs

Recent Models

NAME	TASK	SOURCE	STATUS	CREATED
dice_L_X	DETECT	TRAINING	AVAILABLE	Aug 06, 2026, 08:1
cards 8/6/2026	DETECT	TRAINING	AVAILABLE	Aug 06, 2026, 12:2
cards 8/5/2026	DETECT	TRAINING	AVAILABLE	Aug 06, 2026, 10:3

Recent Benchmarks

NAME	STATUS	PROGRESS	CREATED
QA Compare Seed	COMPLETED	1/1	Aug 06, 2026, 01:28 PM
ts-verify	COMPLETED	1/1	Aug 06, 2026, 11:40 AM
test	COMPLETED	1/1	Aug 06, 2026, 11:19 AM
test1	COMPLETED	1/1	Aug 06, 2026, 10:43 AM

Recent Activity

ACTION	RESOURCE	RESULT	WHEN
AUTH_LOGIN_SUCCEEDED	USER	SUCCESS	Aug 07, 2026, 09:33 AM
AUTH_LOGIN_FAILED	USER	FAILURE	Aug 07, 2026, 09:32 AM
AUTH_LOGIN_SUCCEEDED	USER	SUCCESS	Aug 07, 2026, 09:18 AM
MODEL_CREATED	MODEL	SUCCESS	Aug 06, 2026, 08:12 PM
TRAINING_JOB_COMPLETED	TRAINING_JOB	SUCCESS	Aug 06, 2026, 08:12 PM
AUTH_LOGIN_SUCCEEDED	USER	SUCCESS	Aug 06, 2026, 04:48 PM
USER_CREATED	USER	SUCCESS	Aug 06, 2026, 04:48 PM
TRAINING_JOB_RUNNING	TRAINING_JOB	SUCCESS	Aug 06, 2026, 04:45 PM
TRAINING_JOB_PREPARING	TRAINING_JOB	SUCCESS	Aug 06, 2026, 04:45 PM
TRAINING_JOB_SUBMITTED	TRAINING_JOB	SUCCESS	Aug 06, 2026, 04:45 PM

- **Counters** — Source Datasets, Training Datasets, Models, Training Jobs, Benchmarks
- **System Health** — workers online/offline/total, active executions, pending outbox, dead-letter outbox
- **Active Jobs** — what is running or queued right now
- **Recent Models, Recent Benchmarks, Recent Activity** (latest audit events)

Two banners can appear above any page, for every signed-in user:

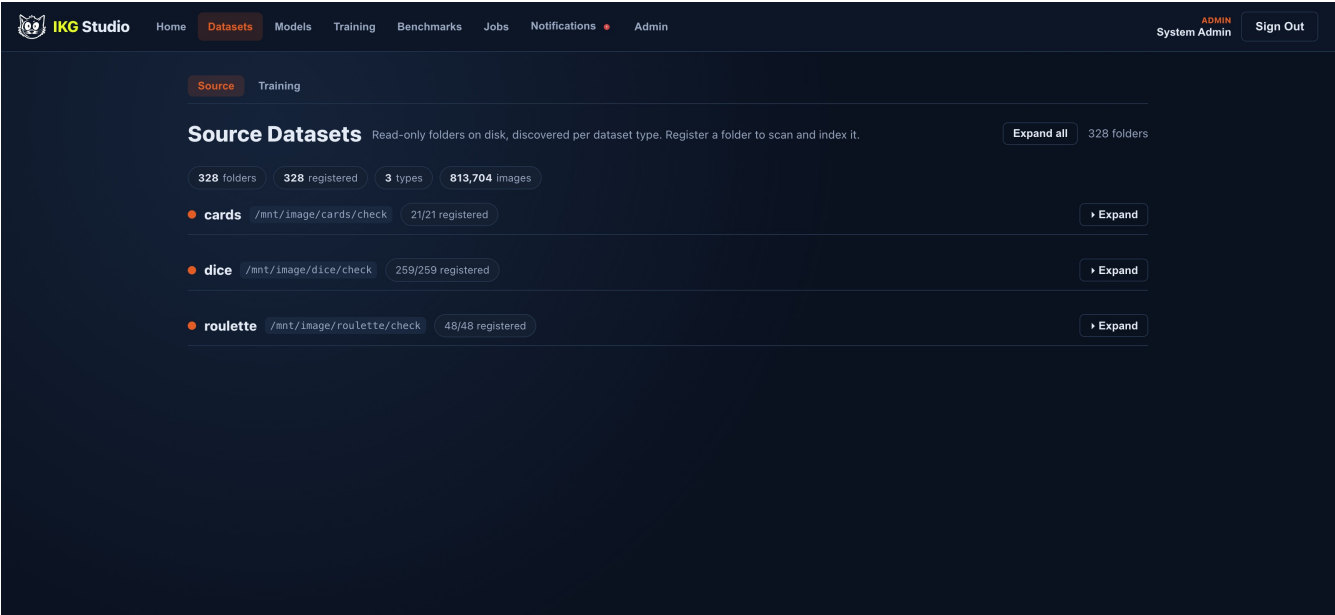
- **Storage Usage Warning** — MinIO usage has passed `storage_warning_threshold_percent` (default 85%).
- **Storage Limit Exceeded** — the quota in `storage_minio_limit_bytes` is full and **uploads and executions are write-blocked**. Free space by deleting unused models, conversions or benchmark artifacts.

6. Datasets

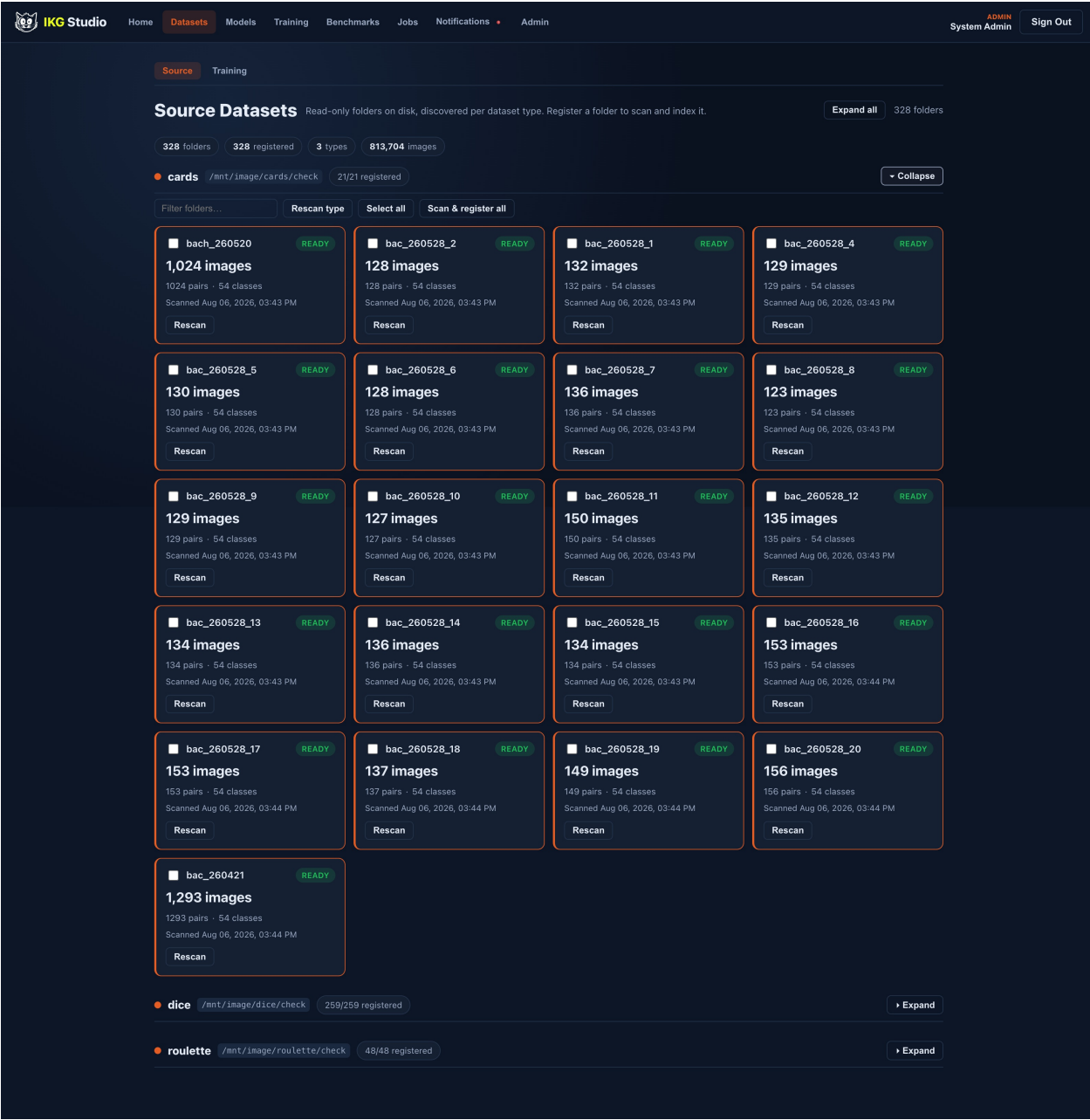
Two sub-tabs: **Source** (raw folders on disk) and **Training** (curated, split datasets).

Both tabs group rows by dataset type in collapsible bands, remember which bands you left open, and offer **Expand all / Collapse all**.

6.1 Source Datasets



Each type band shows a stats strip (folders / registered / types / images) and, when expanded, one card per folder found under the type's dataset path.



Per group: a folder name filter, a multi-select status filter with live counts (`READY` (12) , `Not registered` (5) , ...), **Select all / Deselect all**, **Rescan type** (rebuilds the type's directory index), **Scan & register all** (asks for confirmation above 10 folders), and **Scan & register selected (n)** once you tick a subset.

Per card: scan status, image / pair counts, class count, last-scan time, **Rescan**, and **Archive**.

- A card warning "**⚠ classes.txt missing — using type fallback**" means the folder had no class list and the type's fallback was used.
- A card reading "**⚠ folder not found on disk — open it to archive**" is a registered dataset whose folder has left the directory index (the type's path changed, or the folder moved). **Rescan** recovers it if the folder comes back; otherwise archive it.

Archiving. Archive hides a source dataset from the grid and stops it being used for new training datasets. **Nothing on disk is touched**, and training datasets already built from it are unaffected. Archive is **one-way** — to use the folder again you must register it afresh. Tick several cards to get **Archive selected (n)**; the operation runs one dataset at a time so a partial failure tells you exactly which one failed. Tick **Show archived** in the page header to see a read-only band of archived datasets per type.

Detail page. Clicking a folder opens its detail page:

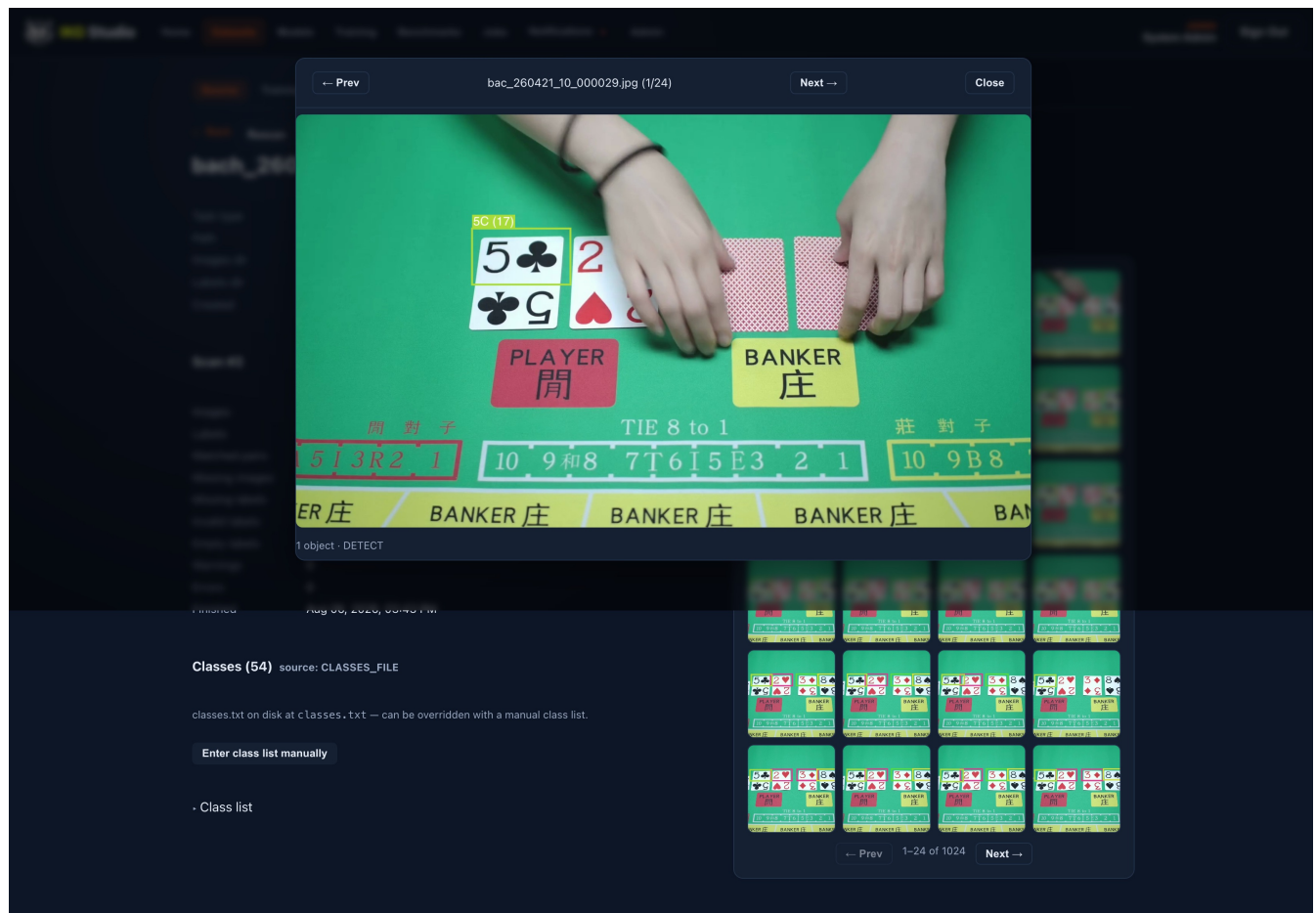
The screenshot shows the IKG Studio interface. At the top, there's a navigation bar with 'Home', 'Datasets', 'Models', 'Training', 'Benchmarks', 'Jobs', 'Notifications', and 'Admin'. The 'Datasets' tab is active. Below the navigation bar, there's a header for the dataset 'bach_260520' with a 'READY' status. The main content area is divided into two columns. The left column contains a table with metadata: Task type (DETECT), Path (/mnt/image/cards/check/bach_260520), Images dir (images), Labels dir (labels), and Created (Aug 05, 2026, 04:58 PM). Below this is a 'Scan #2' section with a table showing statistics: Images (1024), Labels (1024), Matched pairs (1024), Missing images (0), Missing labels (0), Invalid labels (0), Empty labels (0), Warnings (0), Errors (0), and Finished (Aug 06, 2026, 03:43 PM). The right column features a 'Preview' section showing a grid of 1024 small images, each with a status badge (e.g., 'READY', 'INVALID'). At the bottom of the preview grid, there's a pagination control showing '1-24 of 1024'.

It shows the **latest** scan (`Scan #<version>`) — not a history — with:

- a statistics block: images, labels, matched pairs, missing images, missing labels, invalid labels, empty labels, warnings, errors, finished time;
- a **Classes** table (index / name / object count) with a badge saying where the class list came from;
- a **Scan Issues** table (severity, code, image, label, line, details);
- for an `INVALID` dataset, a red banner grouping the errors by code with a plain-English label and an example file and line — e.g. "*Class index gap — label files reference class ids missing from the class list*".

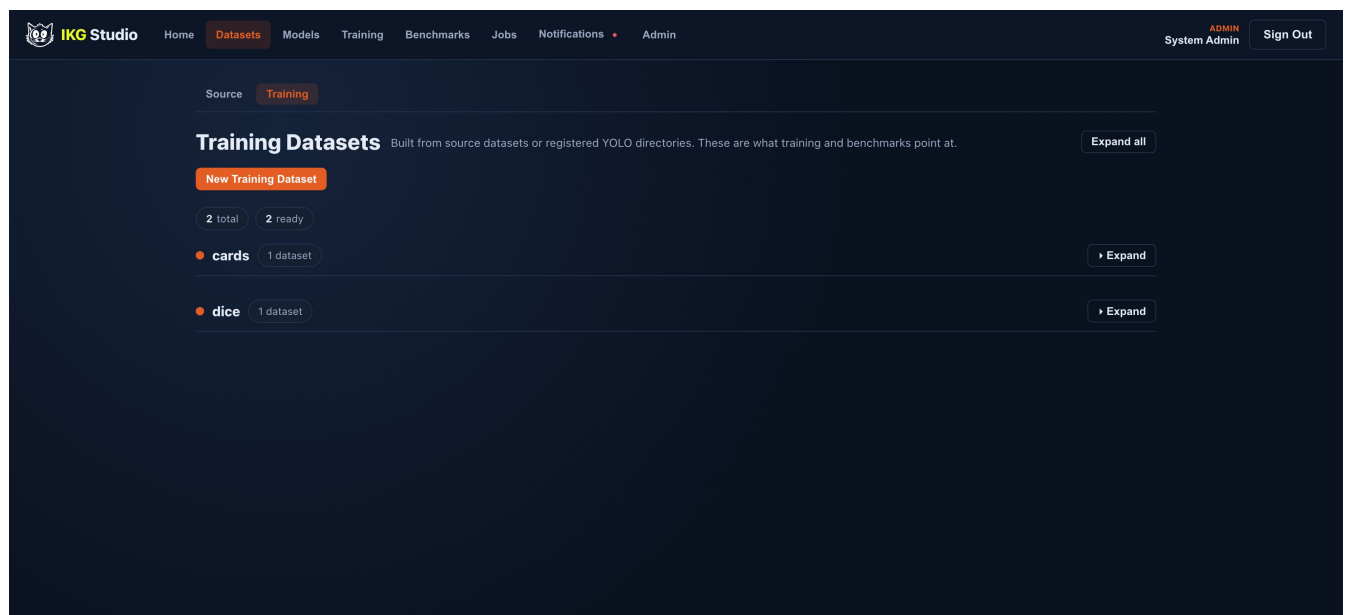
Class list override. If the folder's `classes.txt` is missing or wrong, the detail page can override it: an editor whose line numbers are the class ids, an **Upload classes.txt** picker, and **Save & rescan**, which validates every label file against the new list first. The override is stored on the record and **never written into the source folder**; **Clear override** returns to the on-disk list.

Sample preview. The preview panel renders thumbnails with the boxes already drawn, 24 per page. Click one for the full-size modal:



Modal keys: **← / A** previous, **→ / D** next (wrapping), **Esc** close, and **L** to toggle the class-name tags (there is also a **Labels** button). Both axis-aligned and oriented boxes are drawn, each class in its own colour. Training datasets get **train / val / test** tabs; source datasets are flat.

6.2 Training Datasets



Click an entry for its configuration, class list, split counts and lineage back to the source datasets:

[Home](#)
[Datasets](#)
[Models](#)
[Training](#)
[Benchmarks](#)
[Jobs](#)
[Notifications 1](#)
[Admin](#)

[System Admin](#)
[Sign Out](#)

Source

Training

[← Back](#)

qa_cards_test

READY

Built from source datasets - DETECT - Version 1

Training Dataset

Delete

Origin
Built from source datasets

Task type
DETECT

Path
datasets/qa_cards_test-313e7a3c

Created
Aug 05, 2026, 05:00 PM

Ready
Aug 05, 2026, 05:00 PM

Build Configuration

Split strategy

RANDOM

Storage mode

COPY

Random seed

42

Train ratio

0.80000

Val ratio

0.10000

Test ratio

0.10000

Config hash

cbb15e53eaf9c49...

Build started

Aug 05, 2026, 05:00 PM

Build finished

Aug 05, 2026, 05:00 PM

Dataset Splits

1,152 images total

Train 80%

Val 10%

Test 10%

Train

922 images (80%)

Validation

115 images (10%)

Test

115 images (10%)

Sources

2 datasets

NAME	PATH	TASK
bach_260520	/mnt/image/cards/check/bach_260520	DETECT
bac_260528_2	/mnt/image/cards/check/bac_260528_2	DETECT

Classes (54)

Artifacts

click a supported file to preview

TYPE	FILE	SIZE	CREATED
DATA_YAML	data.yaml	672 B	Aug 05, 2026, 05:00 PM
DATASET_MANIFEST	manifest.json	416 KB	Aug 05, 2026, 05:00 PM

Dataset Preview

images with labels

train

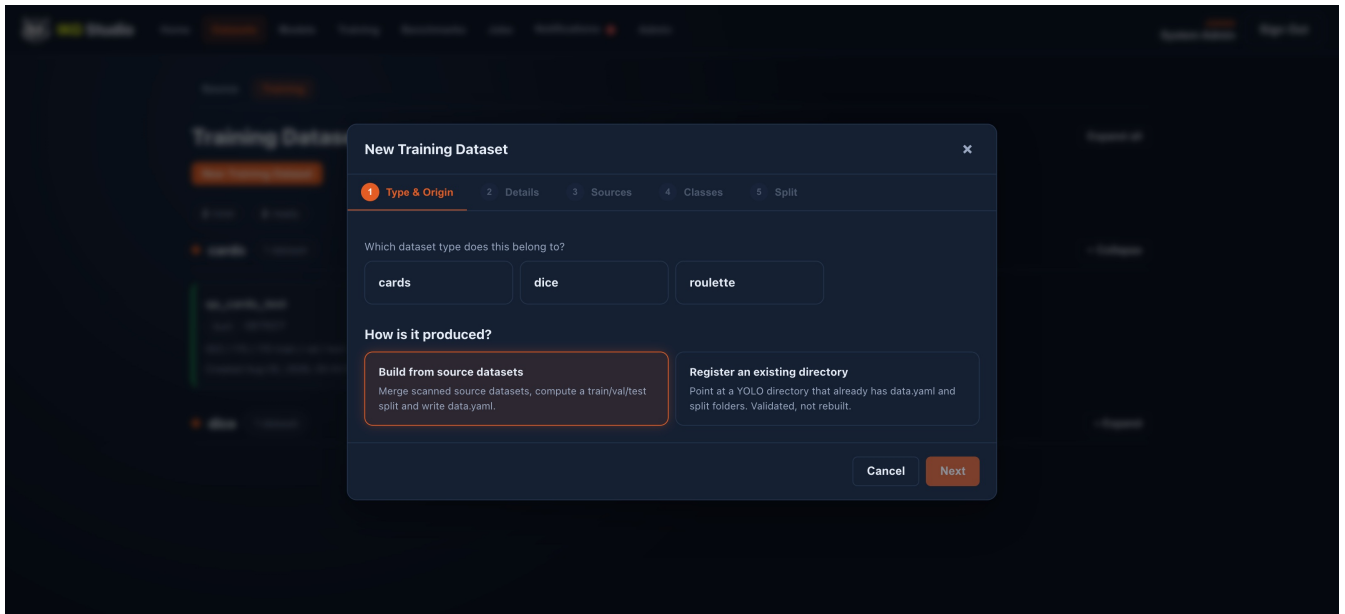
val

test

New Training Dataset. The wizard's length depends on the origin you pick in step 1.

Step 1 — Type & Origin. Choose the dataset type, then one of:

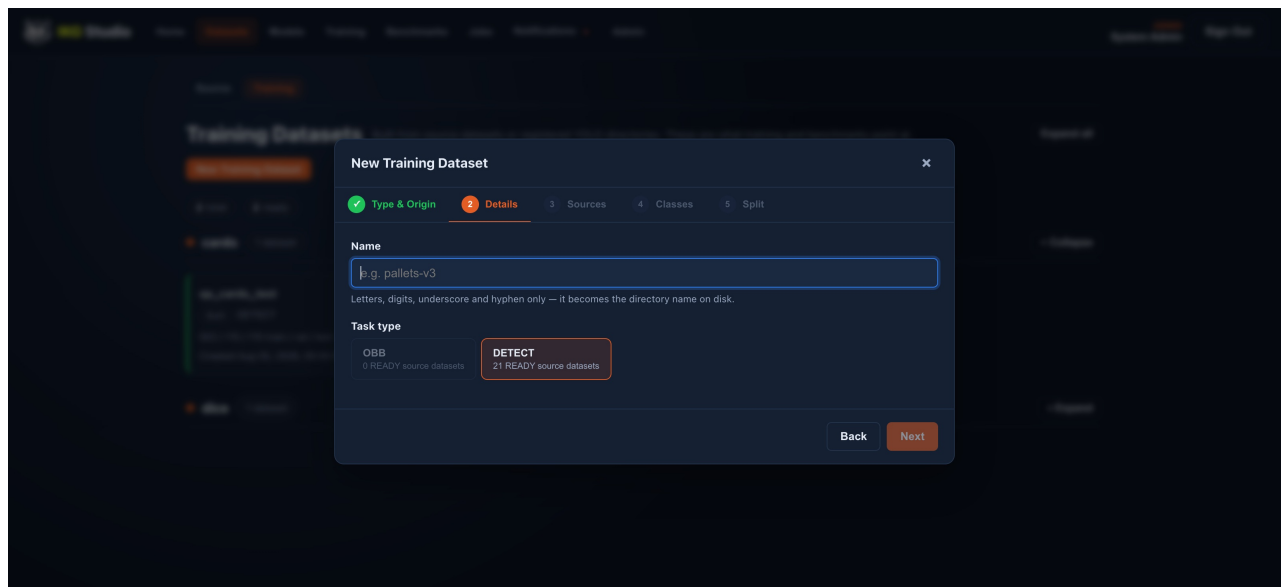
- **Build from source datasets** (BUILT) — merge scanned source datasets, compute a split and write `data.yaml`.
- **Register an existing directory** (REGISTERED) — point at a YOLO directory that already has `data.yaml` and split folders. It is **validated, not rebuilt**.



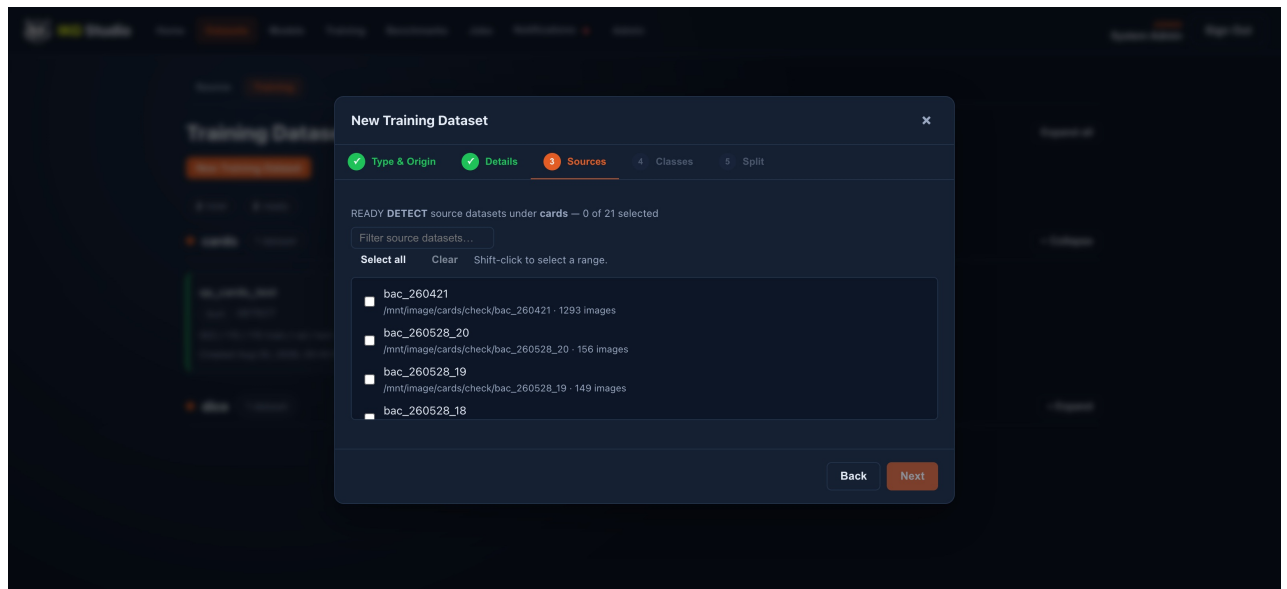
Registering collapses the wizard to three steps — Type & Origin → Details → **Directory** — where you pick the folder with a browser rooted at the type's training-dataset path. The final button reads **Register & Validate**.

Building continues through five steps:

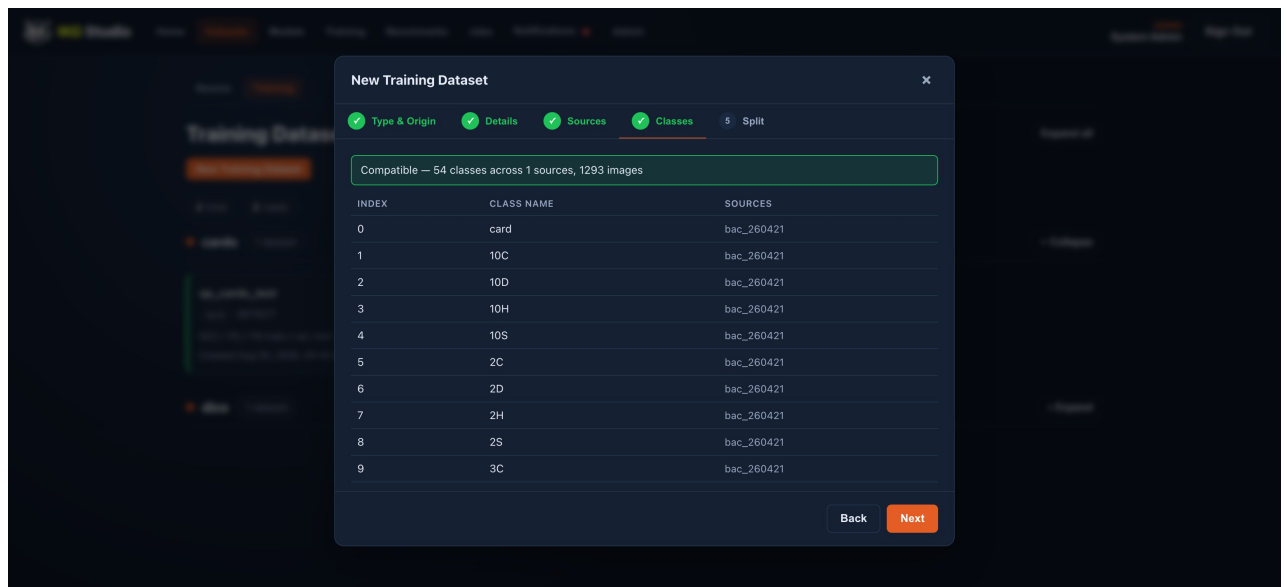
2. **Details** — name and task type (DETECT or OBB). Each task shows a live count of READY source datasets of that task under the type, and is disabled when there are none.



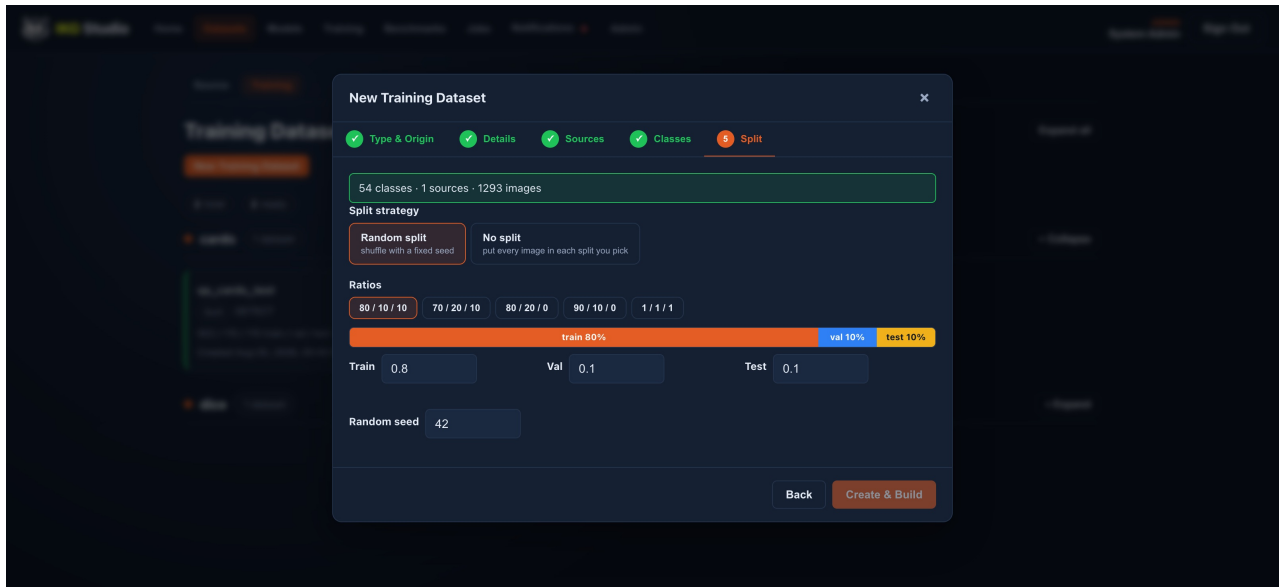
3. **Sources** — pick the source datasets to merge, with a text filter and select all / none. Datasets still scanning are called out as a prerequisite.



4. **Classes** — a **read-only** preview of the merged class list (index / name / which sources contribute it). You cannot rename or deselect classes here. If two sources disagree about what a class index means, the conflict is shown struck-through and **blocks the wizard** — go back and deselect the conflicting source.

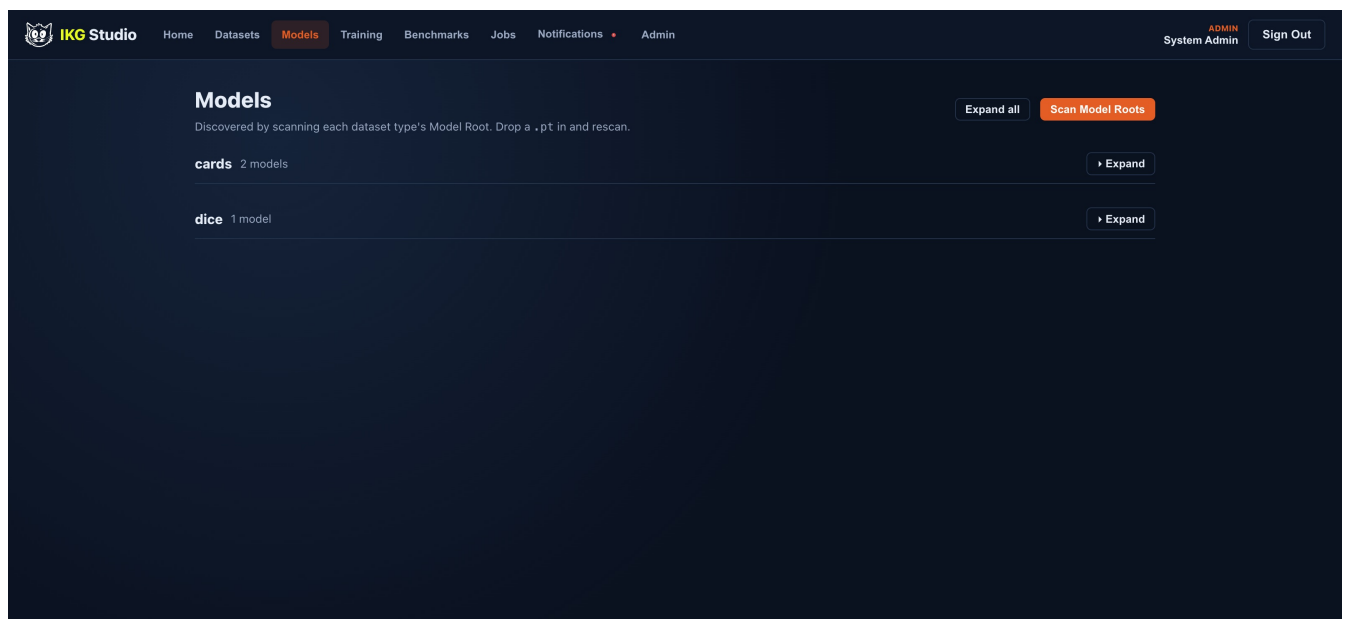


5. **Split** — the strategy and ratios. Two strategies exist:
- **Random split** — shuffle with a fixed seed. Ratio presets (80/10/10, 70/20/10, 80/20/0, 90/10/0, 1/1/1), three editable ratios that must sum to 1.00 with train > 0, a live ratio bar, and a seed field. A warning appears if the train share is very low.
 - **Same split** — keep whatever split the sources already carry. It ignores the ratio fields entirely and must be acknowledged explicitly.



Create & Build queues a **Dataset Build** job; **Register & Validate** queues a **Training Dataset Scan**. Either way, watch it on the [Jobs](#) page.

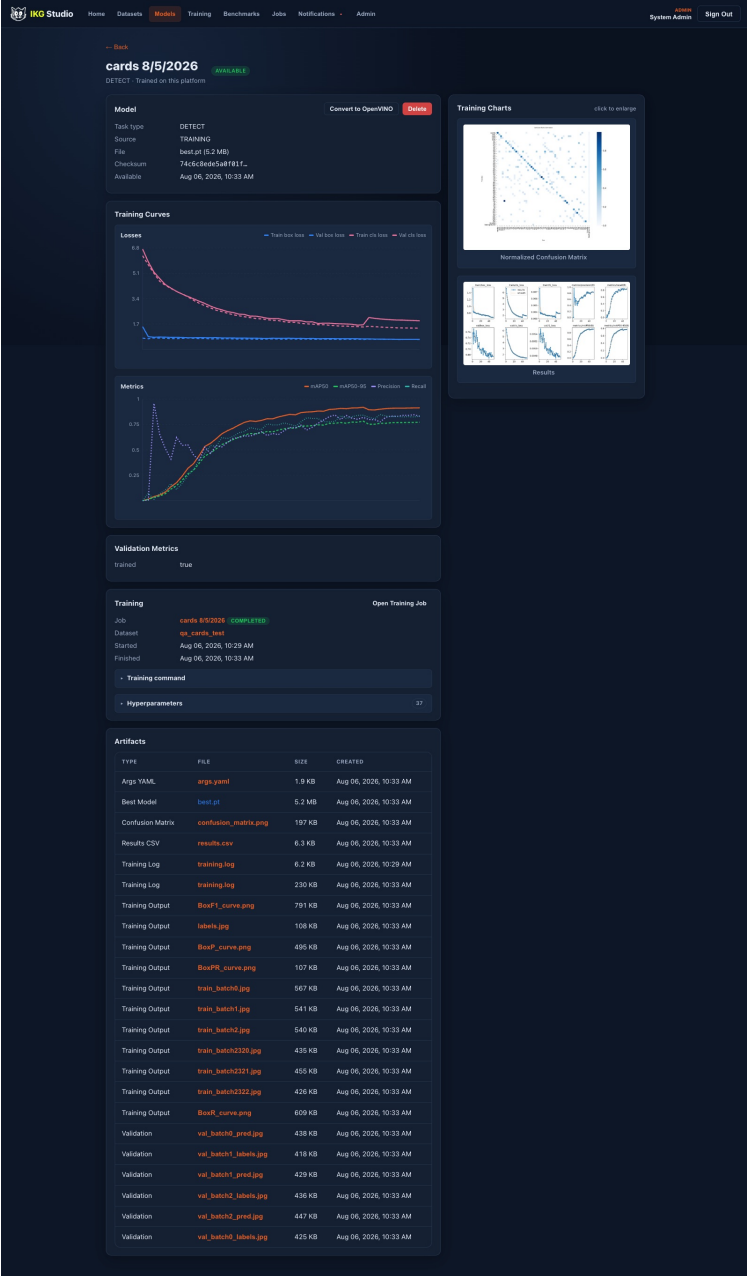
7. Models



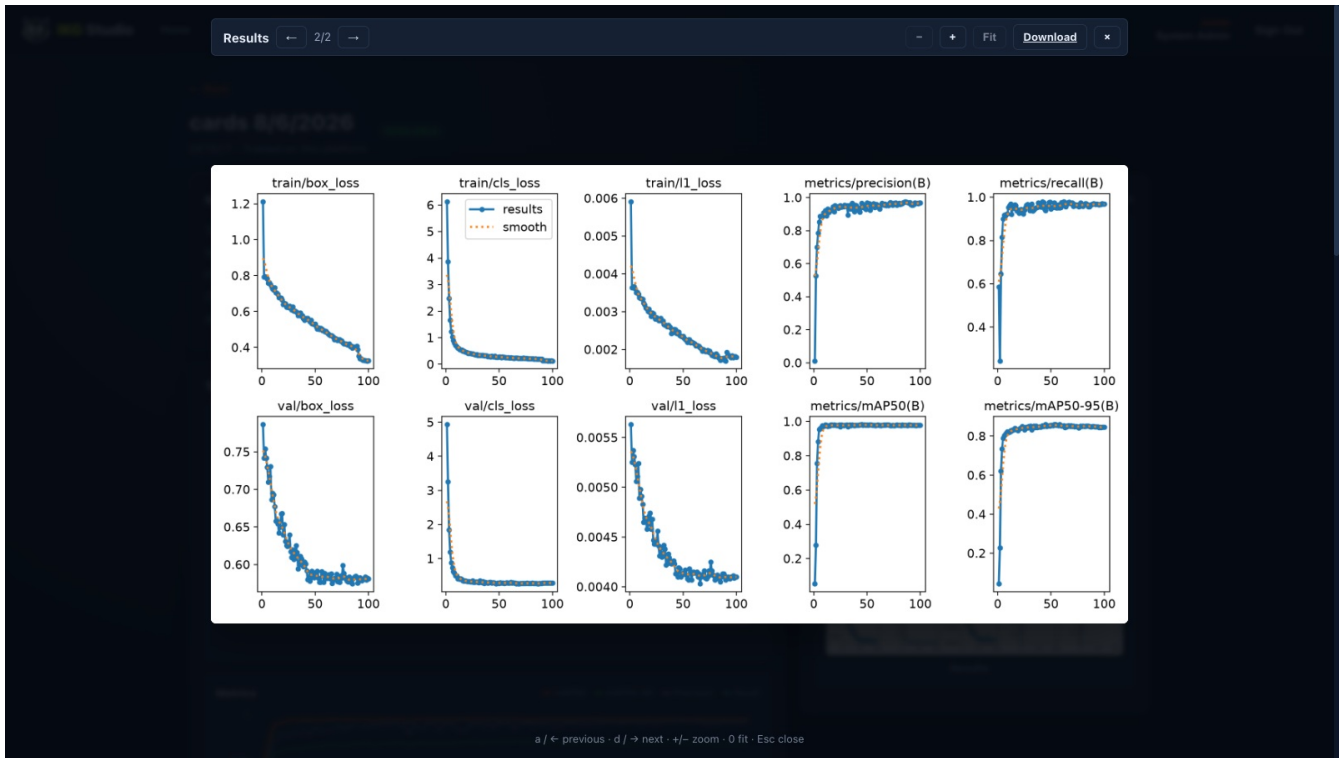
Models are grouped by dataset type. Columns: **Name, Architecture, Task, imgsz, Classes, Status, Size, Added**. A model trained here carries a *Trained here* pill and a `from <base weights>` sub-line.

Scan Model Roots re-indexes each type's Model Root and reports back in plain language — "*N checkpoints found · M newly registered · K Model Root missing on disk*". This is also the **only way to bring an outside model in through the UI**: drop the `.pt` into the type's Model Root and rescan. (The API has URL and file-upload ingest endpoints, and "Model Ingest" appears as a job type, but no screen in this build starts one.)

Model detail shows the file name, size and checksum, training curves, validation metrics, the training job and dataset it came from, a reconstructed `yolo train ...` command line (rebuilt from the stored hyperparameters, with a copy button), the artifacts table, and any OpenVINO conversions.



Click any chart to enlarge it:



Convert to OpenVINO — admin only

Shown only while the model is AVAILABLE. It is a **single modal**, not a wizard:

Convert to OpenVINO

Model: cards_8/6/2026 - DETECT
Output: OpenVINO IR (.xml/.bin) zipped, stored in MiniO

Image Size 3

Image size (imgsz) 640 Non-square imgsz ☐

Export Options 6

Batch 16 (optional) Opset auto (optional)

Max detections 300 Dynamic shape ☐

Simplify ☒ Include NMS ☐

Export CLI Reset

```
yolo export model=cards_8/6/2026.pt format=openvino imgsz=640 simplify=True max_det=300
```

8/6/2026.pt — not a key=value argument

Cancel Convert

- **Image Size** — `imgsz` (32-4096, in steps of 32), with a *Non-square imgsz* toggle that adds a width field.
- **Export Options** — `batch`, `opset`, `max_det` (default 300), `dynamic`, `simplify` (on by default), `nms`.
- An editable **Export CLI** box. Typing in it overrides the form; every `key=value` is validated against the pinned Ultralytics version, with per-argument errors and "did you mean" suggestions. `model` and `format=openvino` are fixed. **INT8 quantisation is not supported** — it needs a calibration dataset.

There is **no device or precision choice**: the conversion runs on the worker's configured device. The output is OpenVINO IR (`.xml` + `.bin`) packed into a single `.zip` artifact in object storage. Nothing is written to the Model Root.

The **OpenVINO Conversions** card below lists every conversion with **Status, Args, Created, Size, Download** and an admin-only delete. It refreshes itself every few seconds while a conversion is queued or running, shows the failure message inline in red when one fails, and clicking a row jumps to that job on the Jobs page.

Delete — admin only

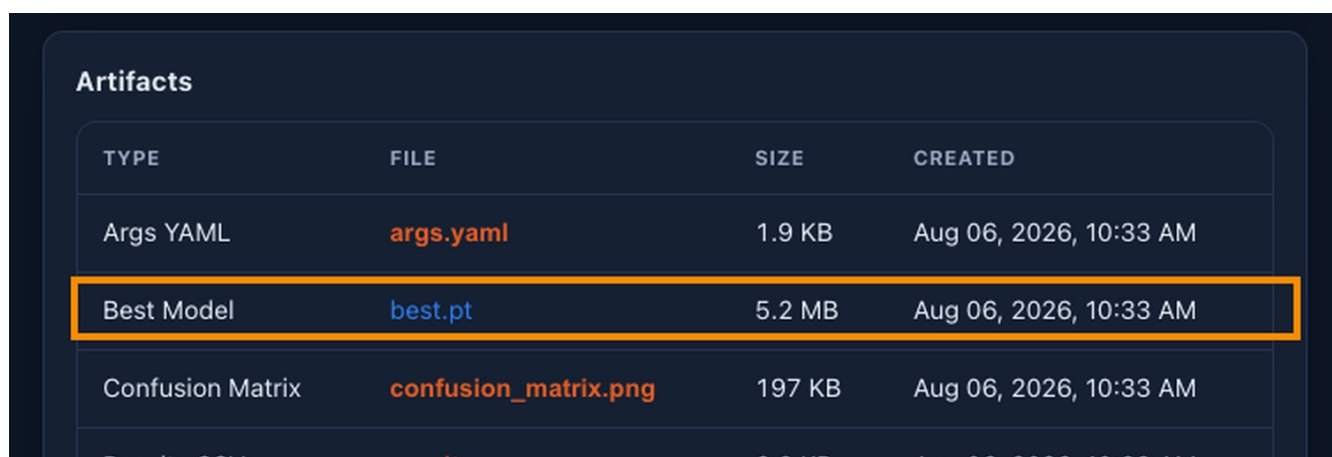
Offered while the model is not already **DELETED**. The confirmation first lists what will be left dangling: training jobs that used it as a base, the training job that produced it, and benchmark runs that reference it. Deletion is a **soft delete** — the record stays with status **DELETED**, the **.pt** is unlinked from the Model Root, and every OpenVINO conversion and its artifact is removed for good. The file removal is fire-and-forget and does **not** appear on the Jobs page.

8. Downloading a Trained Model

This is the question users ask most often, so it gets its own section. Short answer: **Models → open the model → scroll to the Artifacts card → click `best.pt` in the "Best Model" row.**

8.1 From the model page (the normal route)

1. Open **Models** and click the model you want.
2. Scroll to the bottom of the page, past the charts and the Training block, to the **Artifacts** card.
3. Find the row whose **Type** is **Best Model**. Its **File** cell is `best.pt`.
4. Click the file name. The browser downloads it.



The screenshot shows a table titled "Artifacts" with four columns: TYPE, FILE, SIZE, and CREATED. The rows are as follows:

TYPE	FILE	SIZE	CREATED
Args YAML	args.yaml	1.9 KB	Aug 06, 2026, 10:33 AM
Best Model	best.pt	5.2 MB	Aug 06, 2026, 10:33 AM
Confusion Matrix	confusion_matrix.png	197 KB	Aug 06, 2026, 10:33 AM
Results CSV	results.csv	6.2 KB	Aug 06, 2026, 10:33 AM

The row for "Best Model" is highlighted with an orange border.

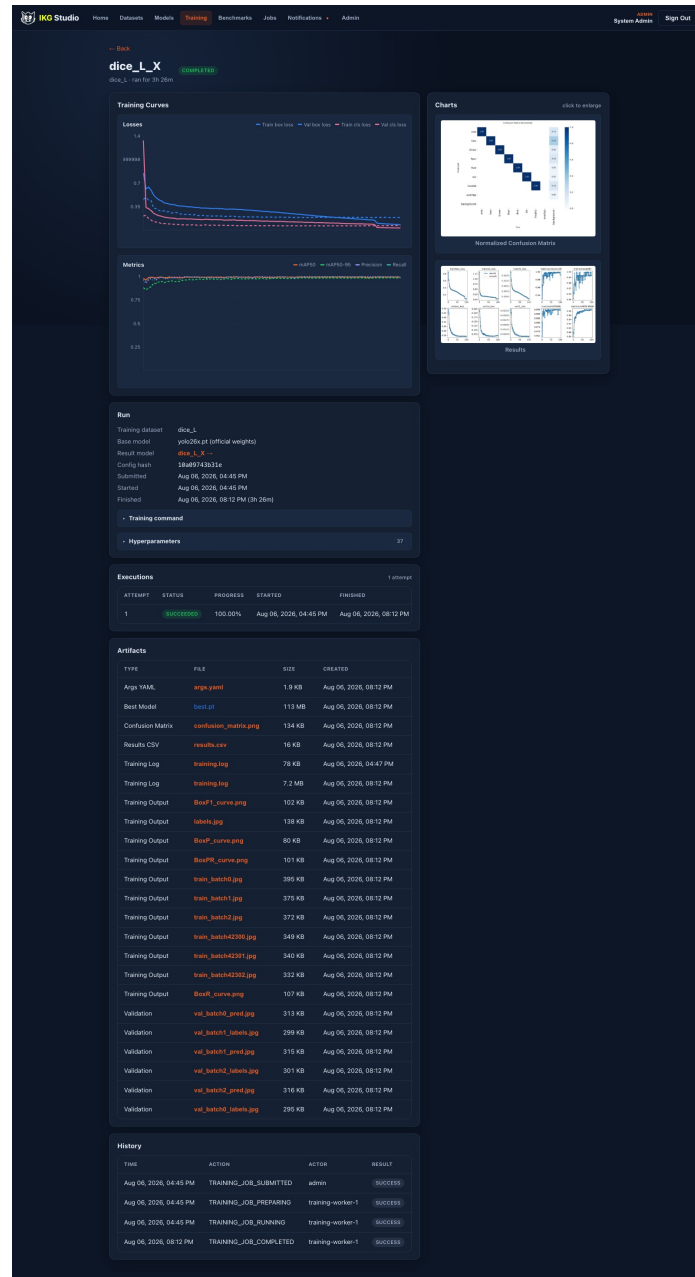
Why it is easy to miss. The card is called *Artifacts*, not *Download*, and there is no download button or icon — the file name itself is the link. Worse, every other row in that table (`args.yaml`, `confusion_matrix.png`, `results.csv`, `training.log`, the batch previews) is a **preview** link and is rendered in orange, while `best.pt` — the one file you actually want — is a plain link in a different colour. Look for the row labelled **Best Model**, not for a button.

The **File** line at the top of the Model card also shows `best.pt` (5.2 MB), but that is a label, not a link.

Both **ADMIN** and **USER** can download artifacts. The file is streamed through the platform (it is not a direct object-storage link), so nothing extra needs to be reachable from your machine.

8.2 From the training job

The same checkpoint is attached to the training run. **Training → the job → Artifacts card → Best Model** gives you the identical file.



The model page hides the job's copy when the model already carries its own, so you see the row once, not twice.

8.3 What else is in that table

Type	File	Use
Best Model	<code>best.pt</code>	The trained weights. This is what you deploy or fine-tune from.
Args YAML	<code>args.yaml</code>	Every hyperparameter the run actually used
Results CSV	<code>results.csv</code>	Per-epoch metrics, for your own plots
Training Log	<code>training.log</code>	Full console output
Confusion Matrix / Training Output / Validation	<code>*.png</code> , <code>*.jpg</code>	Charts and sample batches — these open in a preview, they do not download

Rows that are images or text open in an in-app preview when clicked. Anything else — `best.pt` included — downloads.

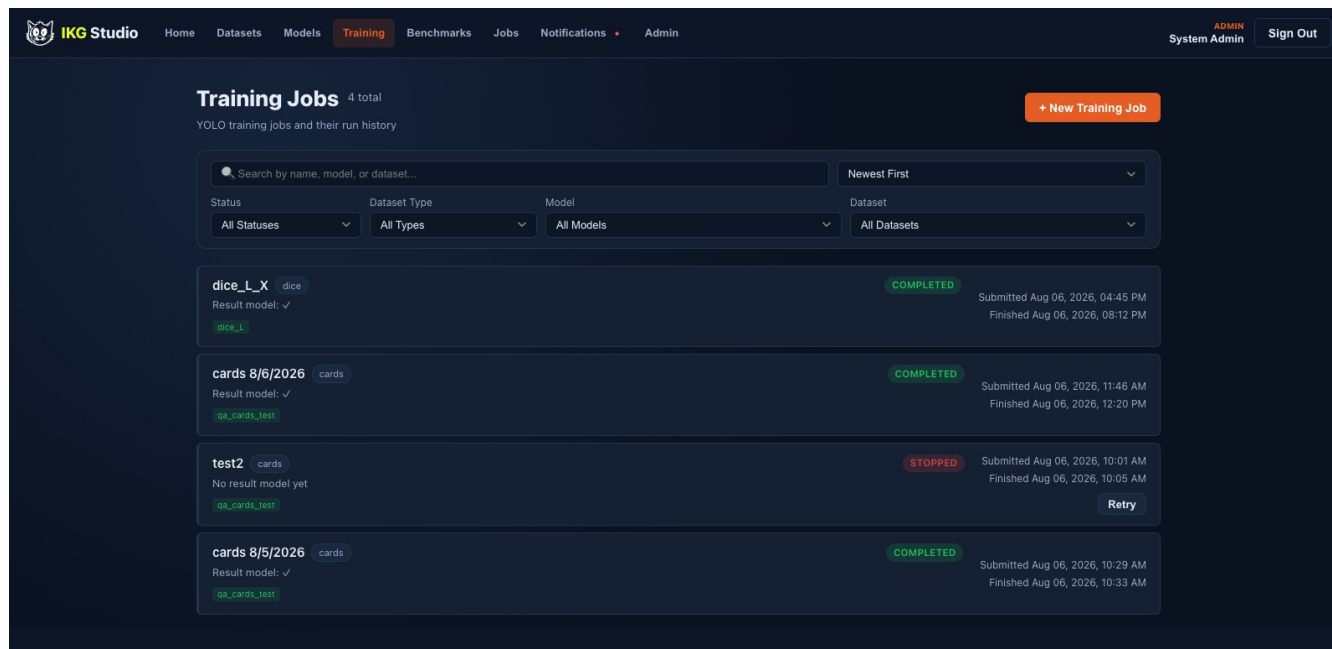
8.4 The OpenVINO export

If you need IR rather than PyTorch weights, use **Convert to OpenVINO** (admin only, [Section 7](#)) and then the **Download** column of the **OpenVINO Conversions** table. That one is labelled "Download". You get a `.zip` containing the `.xml` and `.bin`.

8.5 On the server

Every model trained here is also written as a `.pt` under its dataset type's **Model Root** (Admin → Dataset Types → *model path*). If you have shell access to the server that is the fastest way to collect many checkpoints at once; through the browser, use the Artifacts table.

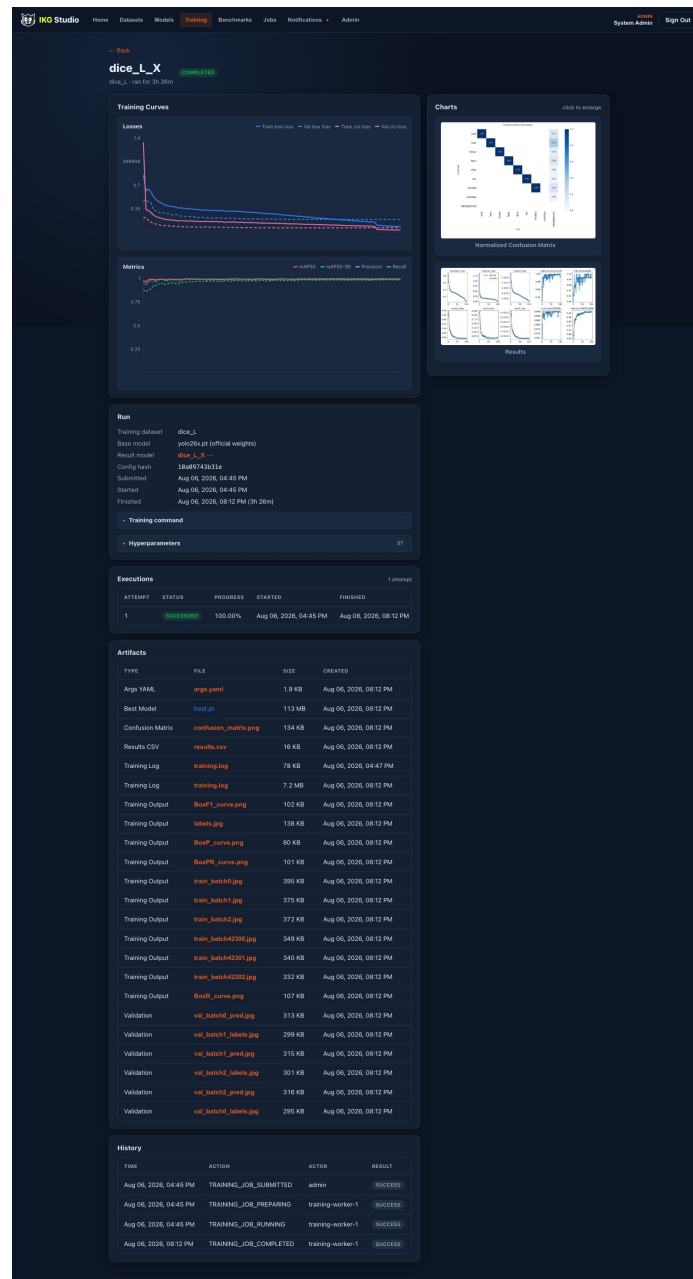
9. Training



The screenshot shows the IKG Studio interface for the Training Jobs page. The top navigation bar includes Home, Datasets, Models, Training (active), Benchmarks, Jobs, Notifications, and Admin. The Training Jobs section shows 4 total jobs. A search bar and filters for Status, Dataset Type, Model, and Dataset are available. The jobs are listed in a table with columns for job name, status, and submission/finish times. The jobs are:

Job Name	Status	Submitted	Finished
dice_L_X	COMPLETED	Aug 06, 2026, 04:45 PM	Aug 06, 2026, 08:12 PM
cards 8/6/2026	COMPLETED	Aug 06, 2026, 11:46 AM	Aug 06, 2026, 12:20 PM
test2	STOPPED	Aug 06, 2026, 10:01 AM	Aug 06, 2026, 10:05 AM
cards 8/5/2026	COMPLETED	Aug 06, 2026, 10:29 AM	Aug 06, 2026, 10:33 AM

The Training page lists every training job. Filters for status, type, model and dataset **cascade** — choosing a model narrows the dataset list and vice versa — plus a sort control (newest / oldest / name) and **Reset filters**. Each row has **Stop** and **Retry** where the status allows it.

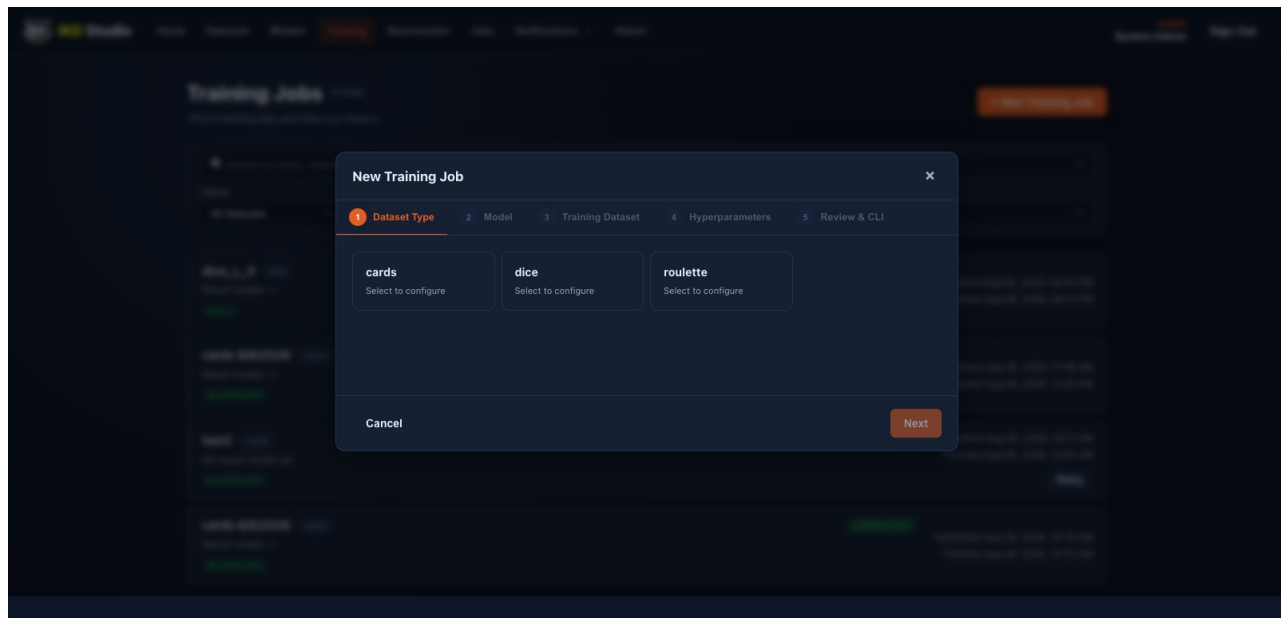


The detail page adds:

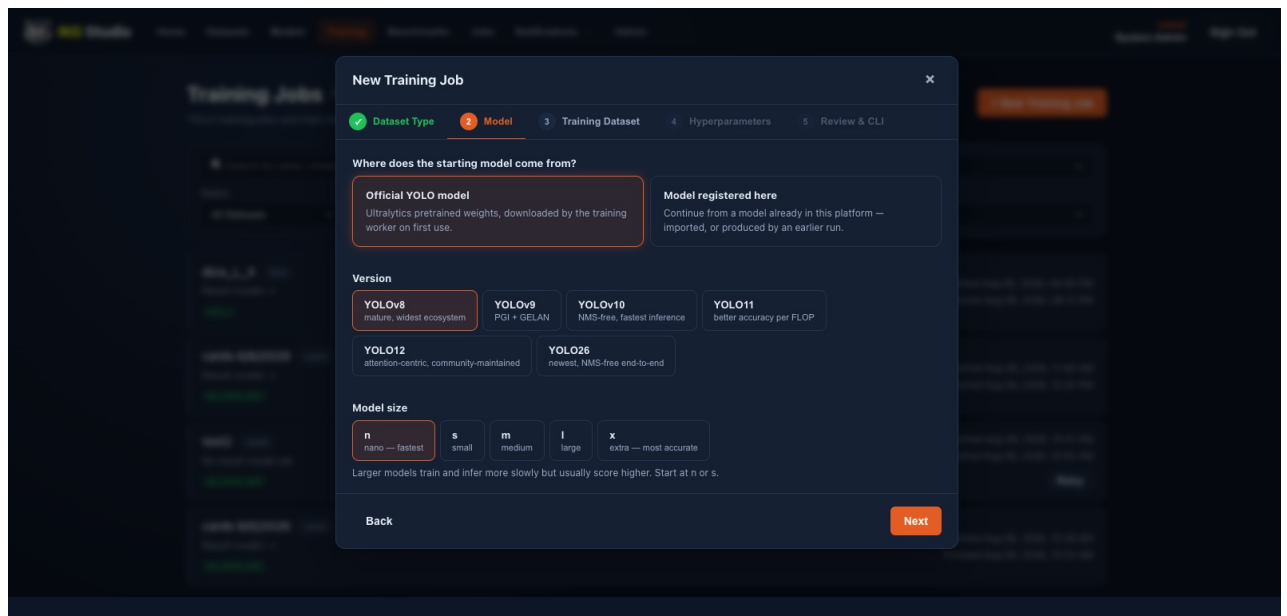
- **Executions** — one row per attempt (Attempt, Status, Progress, Started, Finished, Error), headed "*N* attempts". This is where automatic retries become visible; see [Section 16](#).
- **History** — the audit trail for this job, available to USERS as well as admins.
- Live training curves and a live log tail once the run starts.

New Training Job — five steps

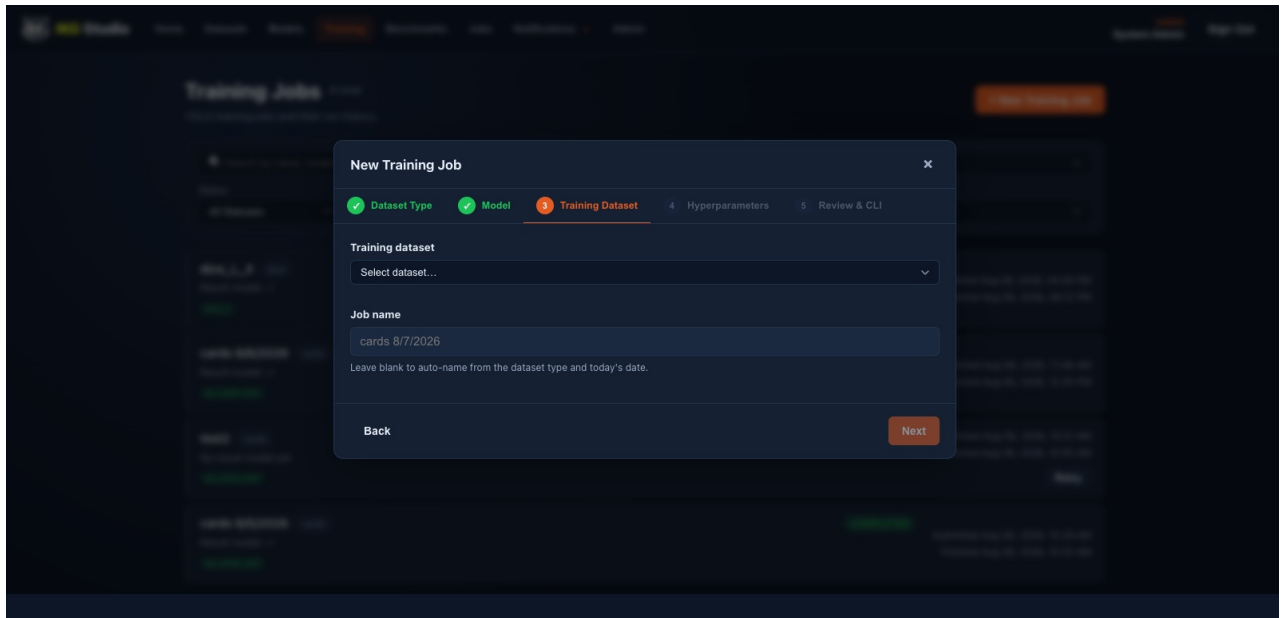
1. **Dataset Type** — everything downstream is scoped to it.



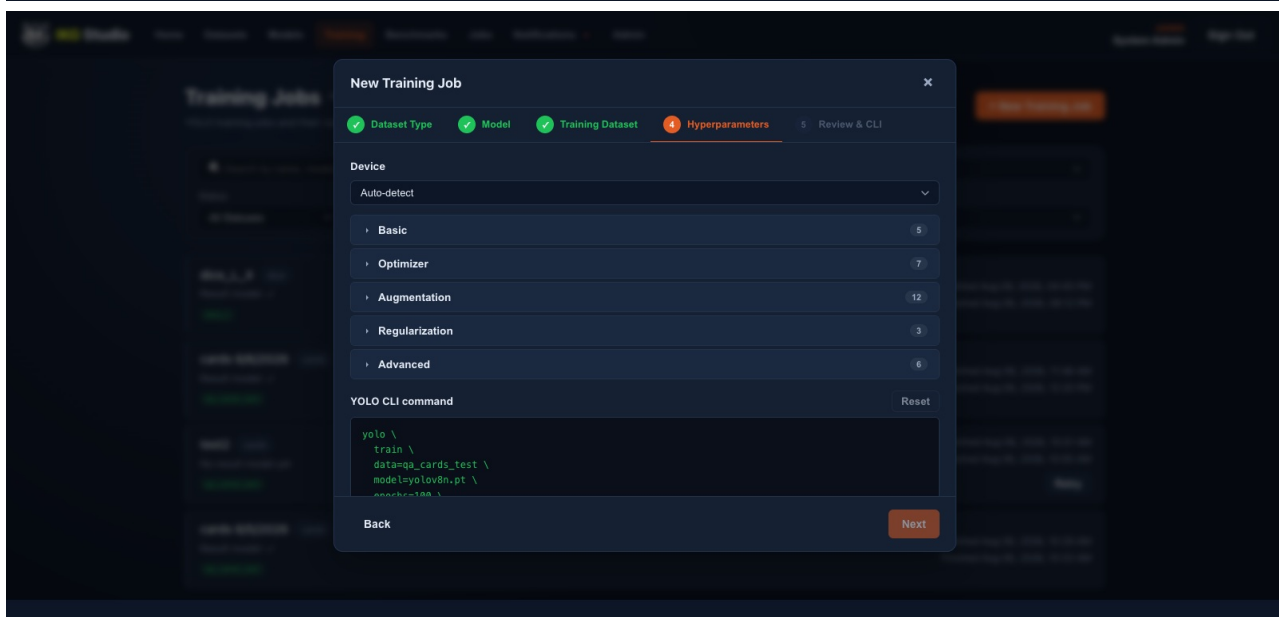
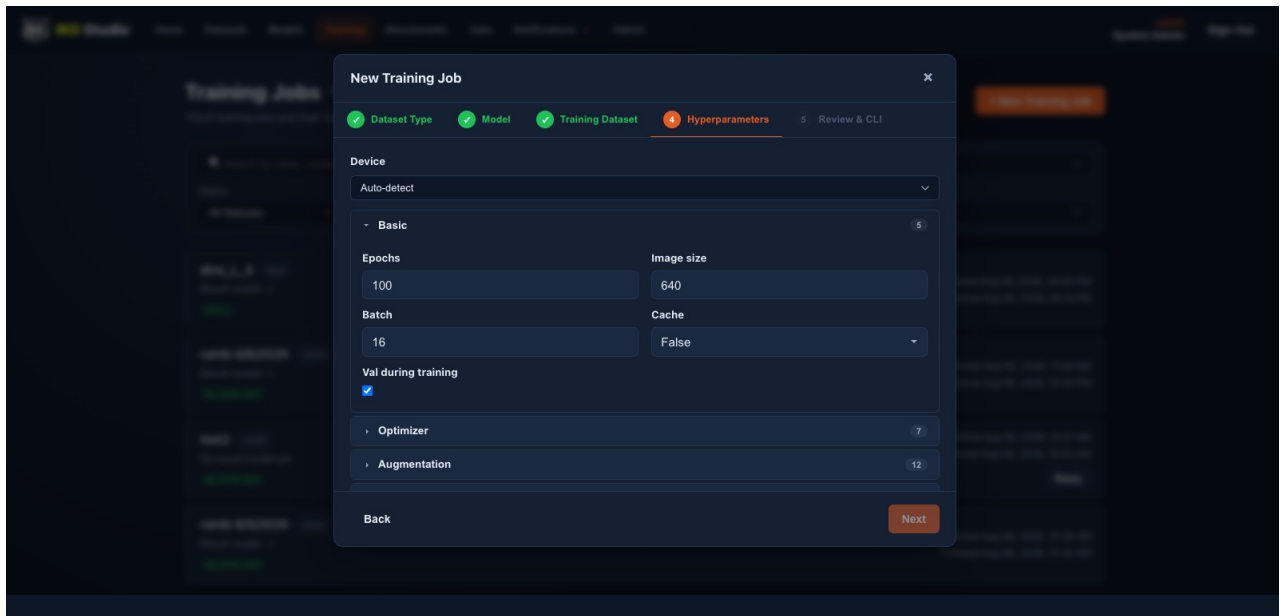
2. **Model** — either **Official YOLO model** (Ultralytics pretrained, fetched by the worker on first use; you choose version and size and the resolved weights file name is previewed) or **Model registered here** (any AVAILABLE model of this type, i.e. fine-tuning).



3. **Training Dataset** — a searchable list of READY datasets with an image and class summary, plus a **job name** (leave blank to auto-name from the type and today's date). Two things block this step: a task-type mismatch between model and dataset, and asking for official OBB weights where that YOLO version has none.

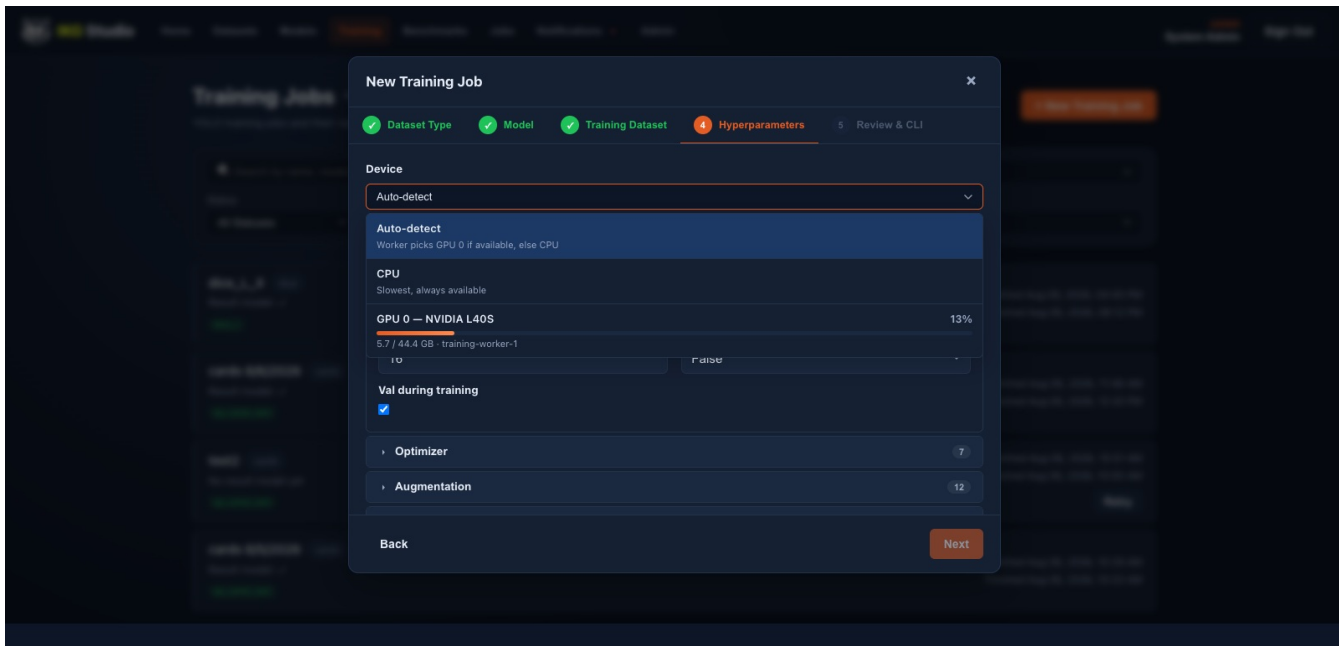


4. **Hyperparameters** — the device picker plus five collapsible groups: **Basic** (epochs, imgsz, batch, cache, val), **Optimizer** (optimizer, lr0, lrf, momentum, weight_decay, warmup_epochs, cos_lr), **Augmentation** (hsv_h/s/v, degrees, translate, scale, shear, flipud, fliplr, mosaic, mixup, copy_paste), **Regularization** (dropout, patience, single_cls) and **Advanced** (workers, seed, save_period, deterministic, multi_scale, rect).



5. **Review & CLI** — the generated `yolo train ...` command, editable. Editing the form regenerates the line; editing the line overrides the form (marked *edited*, with a **Reset**). Arguments are validated against the pinned Ultralytics version.

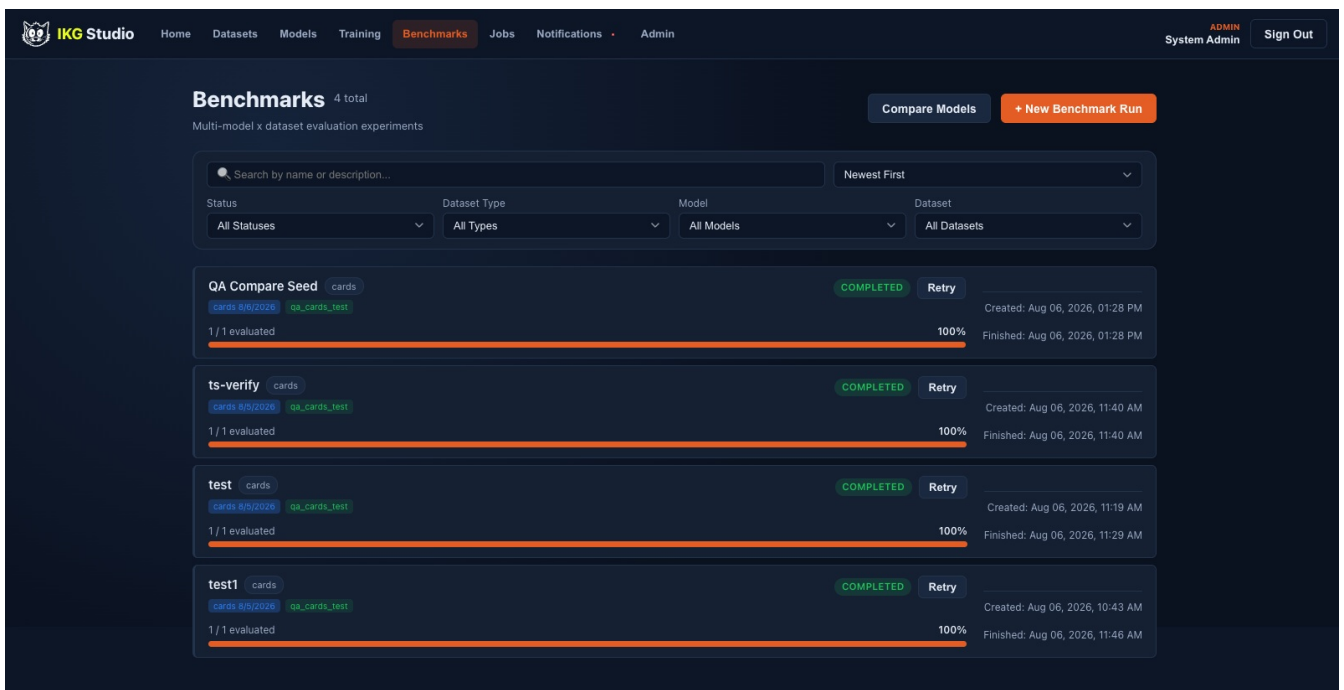
Device picker. Auto-detect, CPU, or one *or several* GPUs — GPU entries are toggles that accumulate into `device=0,1`, and the summary reads "2 GPUs (device=0,1)". Only GPUs reported by **online training workers** are listed, with their live memory and utilisation.



Submitting queues a **Training** job, trackable from [Jobs](#) and [Notifications](#).

10. Benchmarks

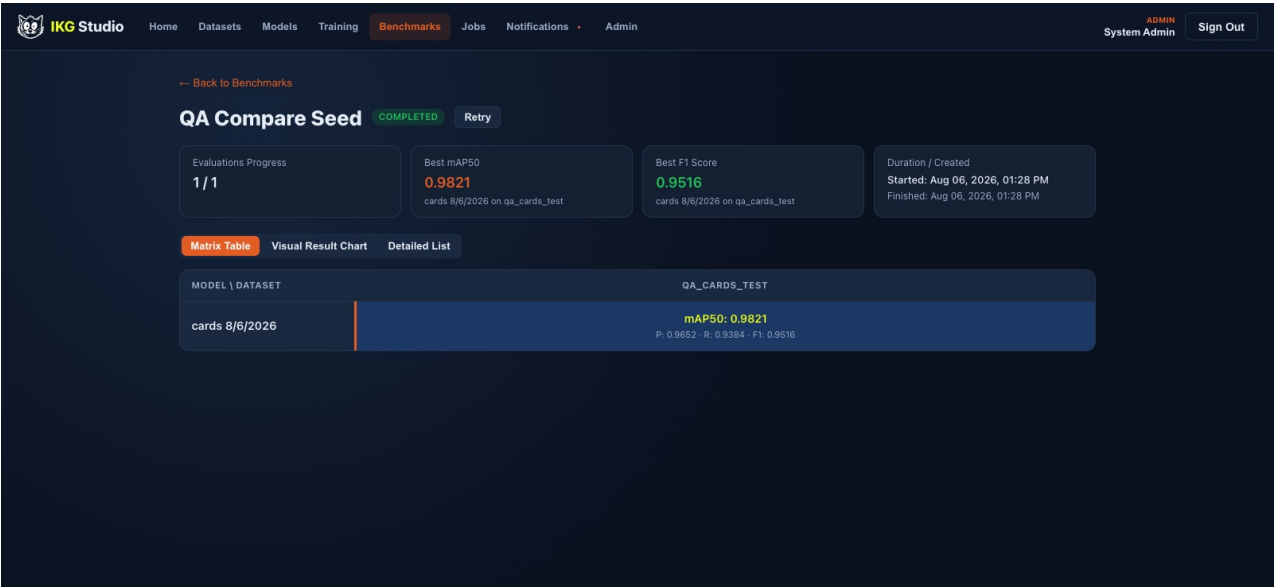
A benchmark run evaluates a model against a training dataset's splits and produces comparable metrics (mAP50, mAP50-95, precision, recall, F1). The underlying schema supports a model x dataset matrix, but the **New Benchmark Run** wizard creates **one model against one dataset** — use **Compare Models** to put several models side by side afterwards.



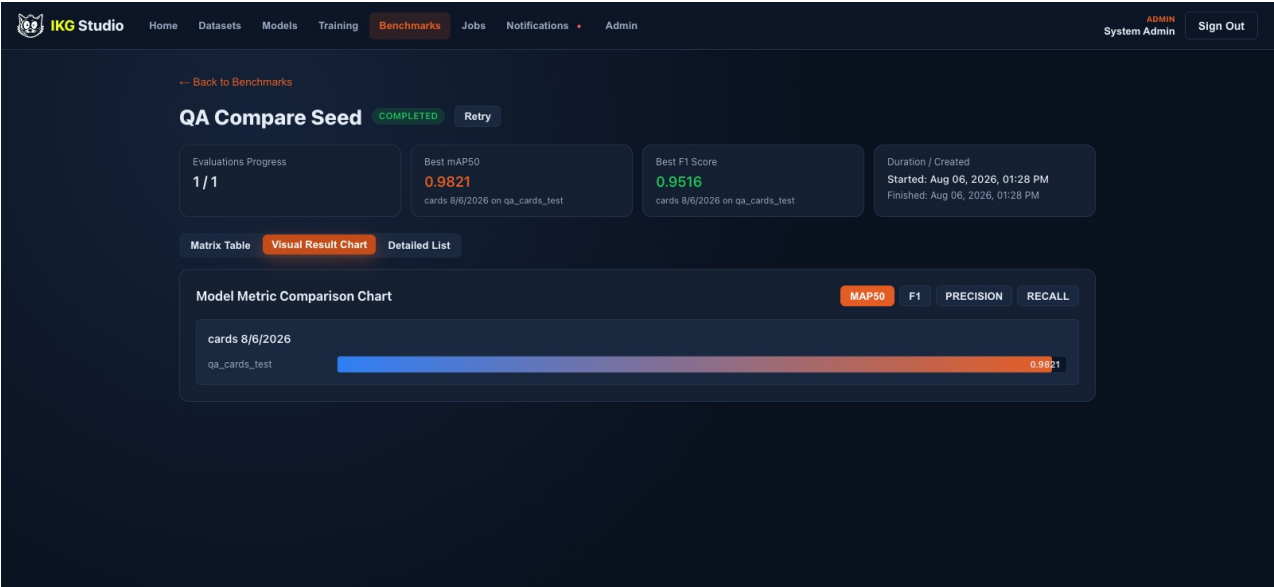
Run detail

Three view modes:

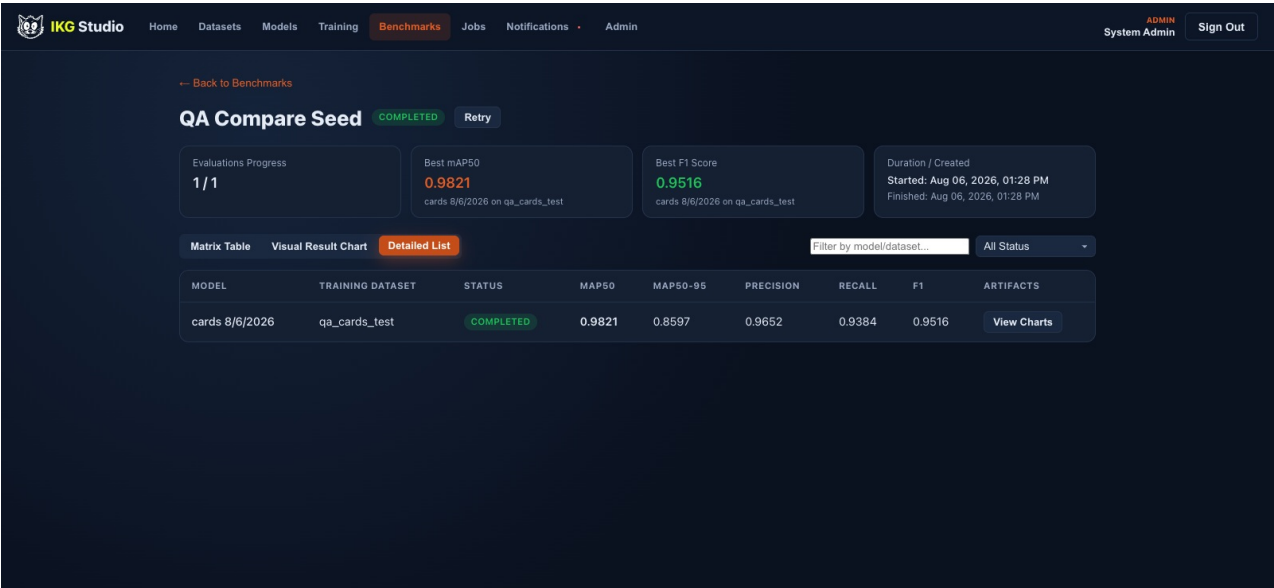
- **Matrix Table** — the model × dataset grid.

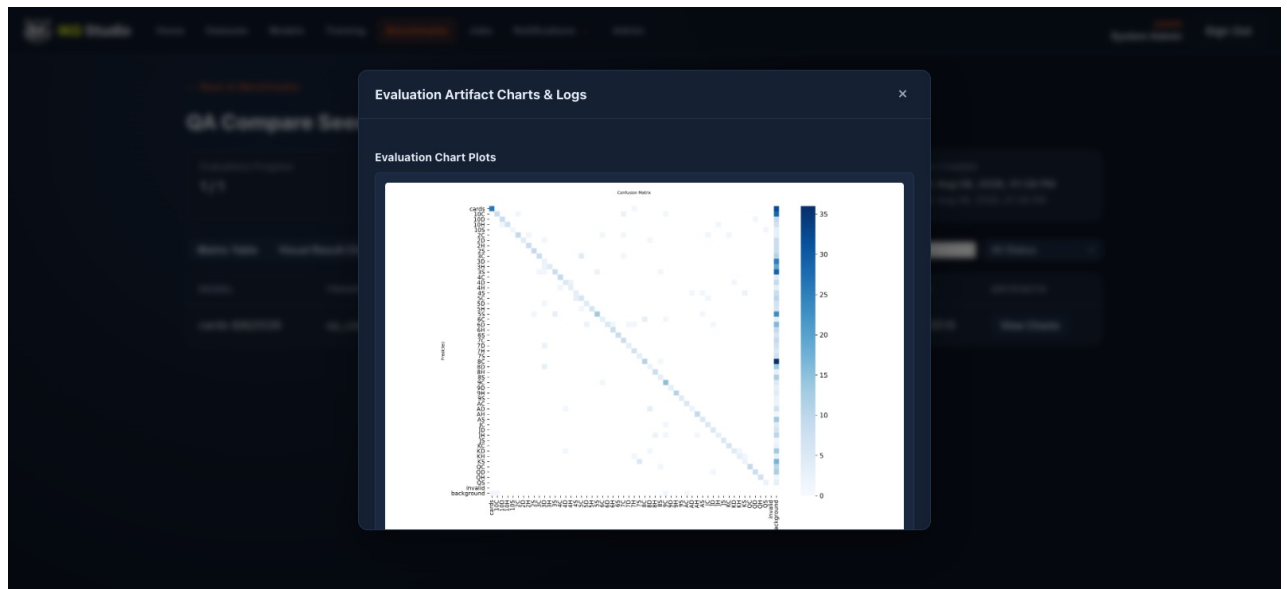


- **Visual Result Chart** — bars comparing mAP50, F1, precision and recall.



- **Detailed List** — one row per evaluation, each with **View Charts** for its confusion matrix, PR curves, per-class charts and logs.





Stop moves a running run to **STOPPING** (queued evaluations are cancelled outright); anything not running or queued refuses.

Retry replaces the whole run. Retry is only offered once a run has finished, and it re-queues **every** evaluation in that run — not just the failed ones — in place, on the same run record. Doing so **clears the existing metrics first**, so retrying a **PARTIALLY_FAILED** run discards the results of the cells that had succeeded and computes them again. There is no per-cell retry. Retry is refused if any model in the run is no longer **AVAILABLE** or any dataset no longer **READY**.

Compare Models

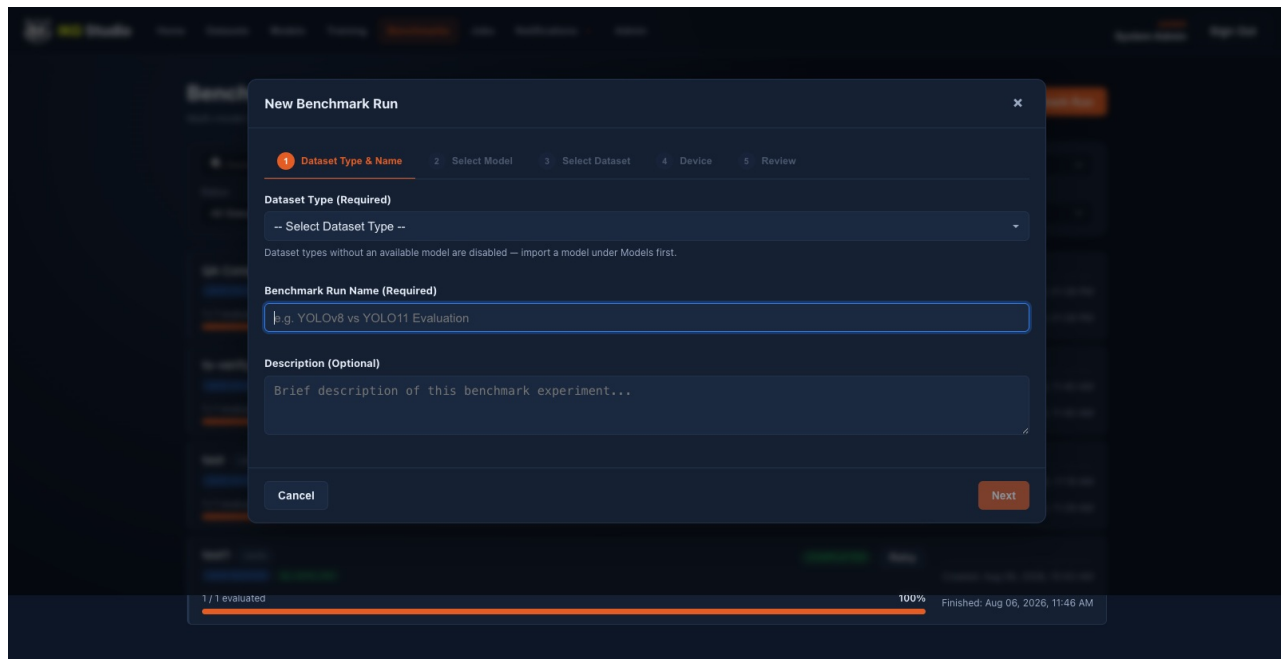
A three-step wizard: **Dataset Type** → **Select Models** → **Compare**. Types with fewer than two models are disabled, and within a type only models that already have a completed evaluation are selectable — the rest are greyed with "*no completed evaluation for this dataset type*".

The image shows a "Compare Models" dialog box with three steps: 1. Dataset Type, 2. Select Models, and 3. Compare. The first step, "Dataset Type (Required)", is active. It features a dropdown menu labeled "-- Select Dataset Type --". Below the dropdown, a message states: "Dataset types with fewer than 2 available models are disabled — nothing to compare yet." At the bottom, there are "Cancel" and "Next" buttons. At the bottom of the main window, a progress bar shows "1 / 1 evaluated" and "100%", with a timestamp "Finished: Aug 06, 2026, 11:46 AM".

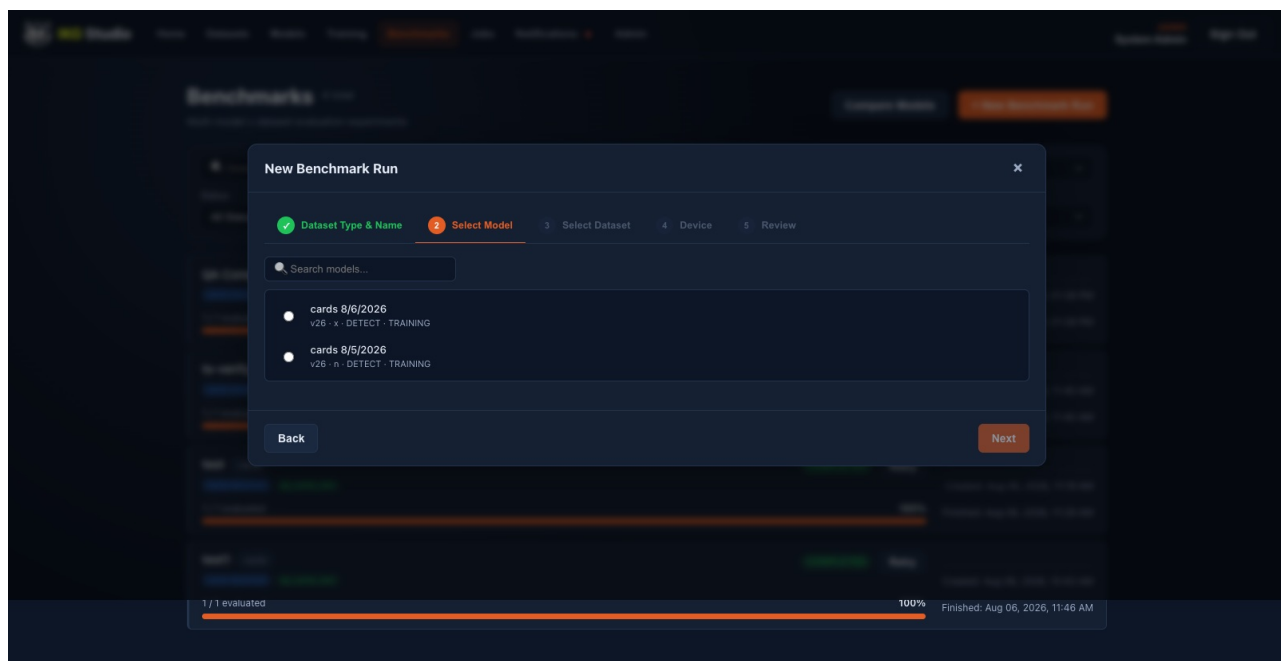
The result view has a **Radar / Bars** toggle, an overlay of the models' training curves, a **↓ CSV** export (overall metrics plus one column per class) and a **↓ PNG** export that works in Radar mode only.

New Benchmark Run — five steps

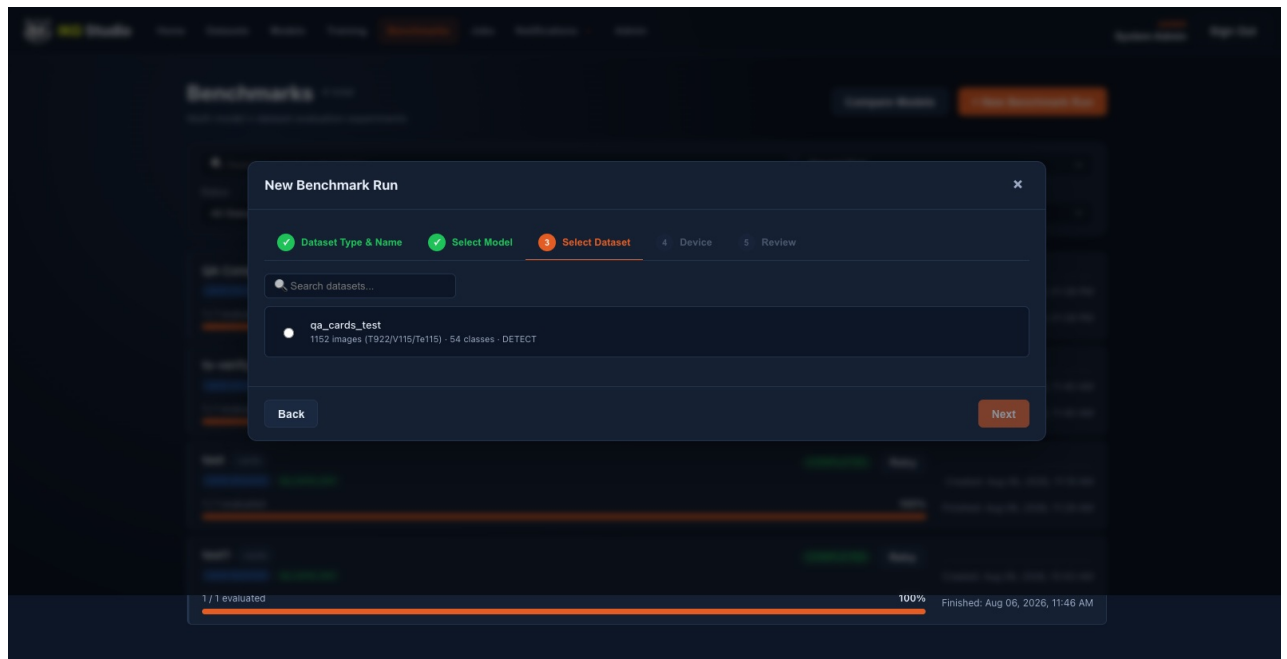
1. Dataset Type & Name



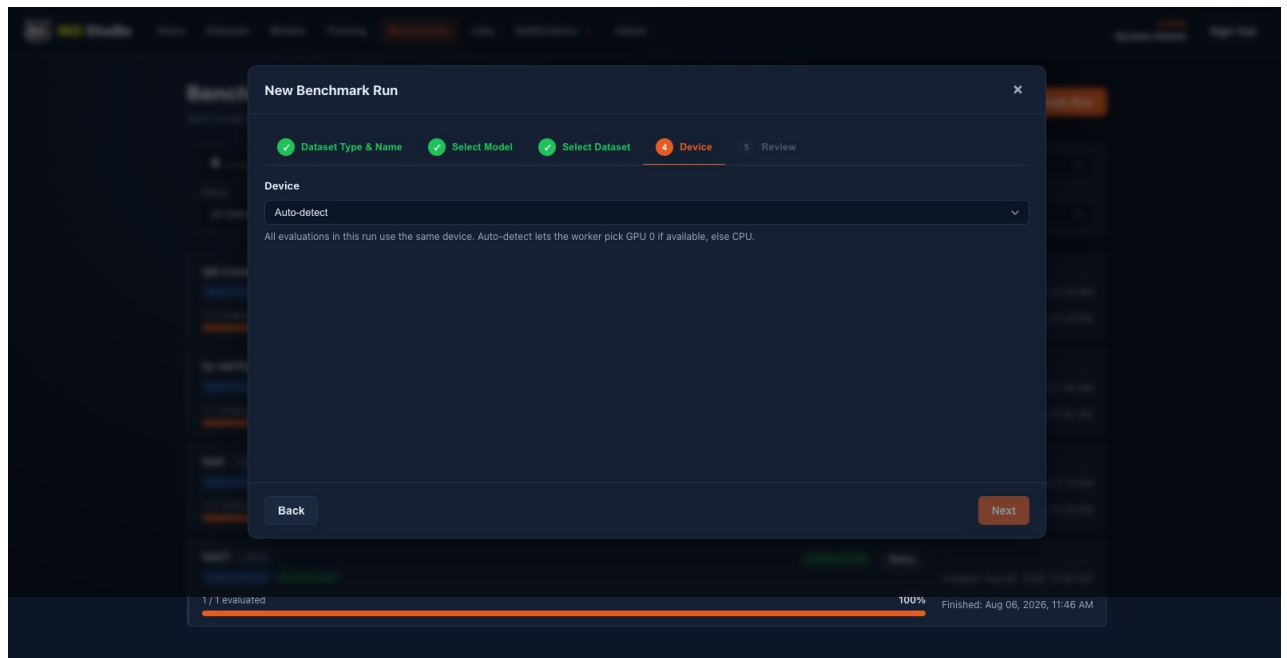
2. **Select Model** — one model.

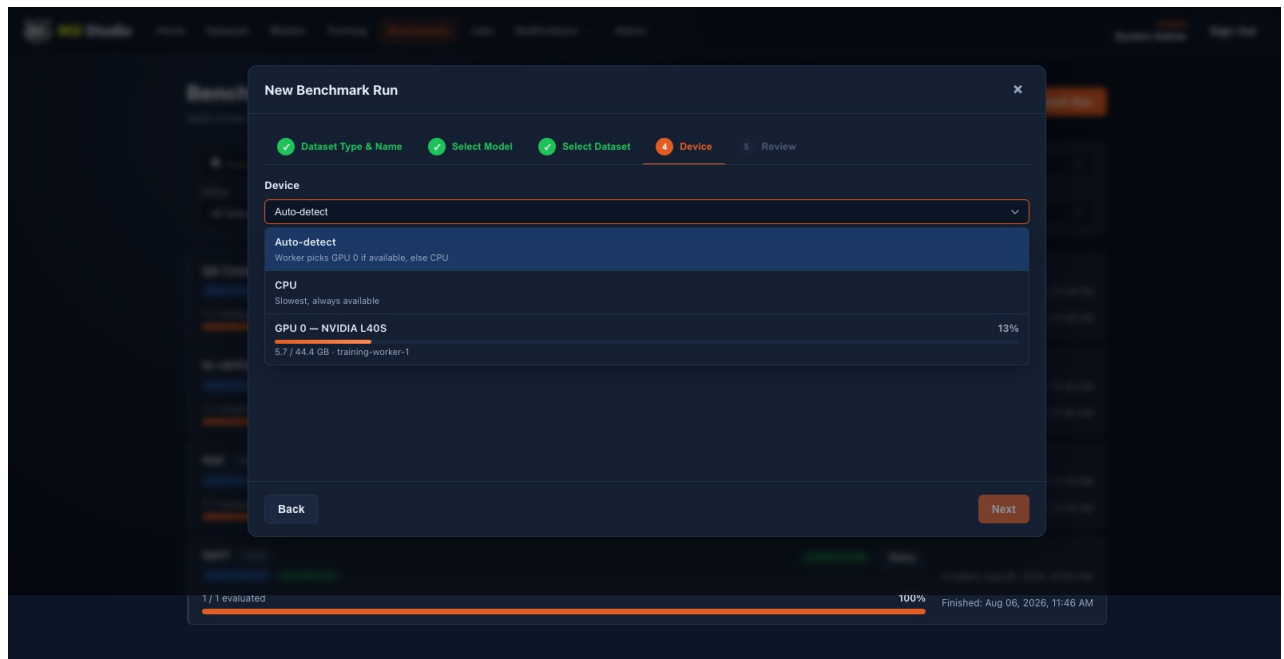


3. **Select Dataset** — one training dataset.

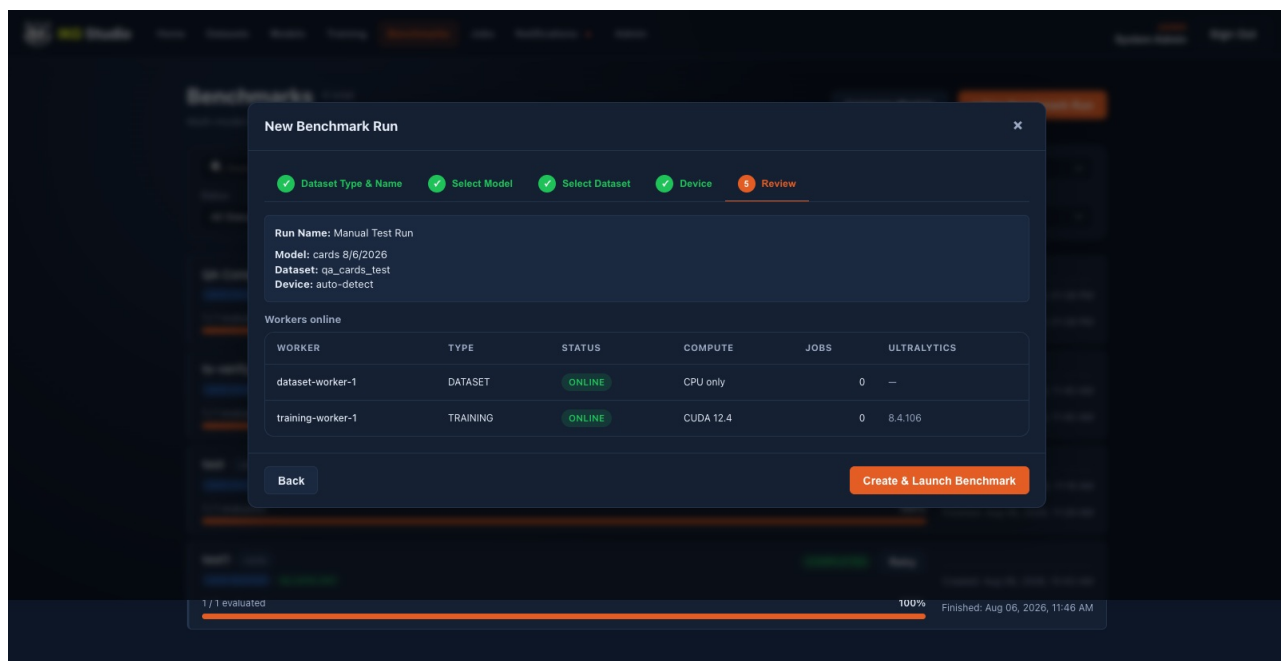


4. **Device** — auto, CPU or a specific GPU, with live worker stats.





5. **Review** — the summary plus a **Workers online** table, then **Create & Launch Benchmark**.



Steps are strictly sequential — you cannot skip ahead.

11. Jobs

Every background job across every subsystem, whichever page launched it.

 IKG Studio

Home

Datasets

Models

Training

Benchmarks

Jobs

Notifications 1

Admin

ADMIN

System Admin

Sign Out

Jobs

415 total

All types

All

Active

Completed

Failed

Training

dice_L_X

SUCCEEDED

(COMPLETED)

100%

3h 26m

2026-08-06 16:45

Build

Build: dice_L

DETECT

SUCCEEDED

(READY)

100%

<1s

2026-08-06 16:38

Build

Build: dice_L

DETECT

LOST

(READY)

46%

5m 0s

2026-08-06 16:24

Scan

Scan: asb12_260106

DETECT

SUCCEEDED

(COMPLETED)

100%

1m 19s

2026-08-06 15:53

Scan

Scan: asb12_260105

DETECT

SUCCEEDED

(COMPLETED)

100%

54s

2026-08-06 15:53

Scan

Scan: asb12_260103

DETECT

SUCCEEDED

(COMPLETED)

100%

27s

2026-08-06 15:53

Scan

Scan: asb12_260104

DETECT

SUCCEEDED

(COMPLETED)

100%

<1s

2026-08-06 15:53

Job types: **Training, Dataset Build, Dataset Validate, Dataset Scan, Benchmark Eval, Model Ingest, Model Conversion**. Quick filters **All / Active / Completed / Failed** sit alongside a type filter; both are remembered across reloads. The list refreshes every few seconds and loads more as you scroll.

 IKG Studio

Home

Datasets

Models

Training

Benchmarks

Jobs

Notifications 1

Admin

ADMIN

System Admin

Sign Out

Jobs

415 total

All types

All

Active

Completed

Failed

All types

Training

Dataset Build

Dataset Validate

Dataset Scan

Benchmark Eval

Model Ingest

Model Conversion

Scan

Scan: asb12_260106

DETECT

SUCCEEDED

(COMPLETED)

100%

3h 26m

2026-08-06 16:45

Scan

Scan: asb12_260105

DETECT

SUCCEEDED

(COMPLETED)

100%

54s

2026-08-06 15:53

Scan

Scan: asb12_260103

DETECT

SUCCEEDED

(COMPLETED)

100%

27s

2026-08-06 15:53

Scan

Scan: asb12_260104

DETECT

SUCCEEDED

(COMPLETED)

100%

<1s

2026-08-06 15:53

Each row shows the job kind, name, **technical status** (the execution's own state) with the **business status** in parentheses (the state of the thing it was working on), progress, duration and timestamp. See [Status Reference](#).

Clicking a row opens a **modal** with the run's log. While the execution is active the log re-polls every few seconds and follows the tail; scroll up and it pauses following until you scroll back down.

Jobs

Dataset Build

Build: dice_L

LOST

Business: READY

46%

Created

8/6/2026, 4:24:35 PM

Started

8/6/2026, 4:24:35 PM

Duration

5m 0s

Finished

8/6/2026, 4:29:35 PM

Log

Materialising item 2581/9398

Execution lost: no heartbeat within timeout

Scan: asb12_251225

DETECT

SUCCEEDED

(COMPLETED)

4m 47s

2026-08-06 15:48

Scan: asb12_251223

DETECT

SUCCEEDED

(COMPLETED)

4m 34s

2026-08-06 15:48

Scan: asb12_251224

DETECT

SUCCEEDED

(COMPLETED)

4m 19s

2026-08-06 15:48

Scan: asb11_260125

DETECT

SUCCEEDED

(COMPLETED)

3m 56s

2026-08-06 15:48

Scan: asb12_251230

DETECT

SUCCEEDED

(COMPLETED)

3m 45s

2026-08-06 15:47

Scan: asb12_251231

DETECT

SUCCEEDED

(COMPLETED)

3m 37s

2026-08-06 15:47

Scan: asb12_260101

DETECT

SUCCEEDED

(COMPLETED)

2m 46s

2026-08-06 15:47

Scan: asb12_260102

DETECT

SUCCEEDED

(COMPLETED)

2m 18s

2026-08-06 15:47

Scan: asb12_251229

DETECT

SUCCEEDED

(COMPLETED)

2m 5s

2026-08-06 15:47

Scan: asb12_251228

DETECT

SUCCEEDED

(COMPLETED)

1m 49s

2026-08-06 15:47

Scan: asb12_251227

DETECT

SUCCEEDED

(COMPLETED)

1m 33s

2026-08-06 15:47

Scan: asb12_251226

DETECT

SUCCEEDED

(COMPLETED)

1m 23s

2026-08-06 15:47

Scan: asb11_260113

DETECT

SUCCEEDED

(COMPLETED)

1m 58s

2026-08-06 15:46

Scan: asb11_260116

DETECT

SUCCEEDED

(COMPLETED)

1m 52s

2026-08-06 15:46

Scan: asb11_260117

DETECT

SUCCEEDED

(COMPLETED)

1m 36s

2026-08-06 15:46

Scan: asb11_260118

DETECT

SUCCEEDED

(COMPLETED)

1m 15s

2026-08-06 15:46

Scan: asb11_260109

DETECT

SUCCEEDED

(COMPLETED)

1m 1s

2026-08-06 15:46

Scan: asb11_260107

DETECT

SUCCEEDED

(COMPLETED)

53s

2026-08-06 15:46

Model deletion and Model Root scans are fire-and-forget and never appear here.

12. Notifications

System events raised per user — job completions, failures, scan problems. The nav badge shows the unread count and updates without a refresh.

[Home](#)
[Datasets](#)
[Models](#)
[Training](#)
[Benchmarks](#)
[Jobs](#)

Notifications

[Admin](#)

ADMIN
System Admin

Sign Out

Notifications

193 total

Mark all read

☐ Unread only

All severities

✓

Training Completed

Training "dice_L_X" completed; model registered.

Aug 06, 2026, 08:12 PM

Mark read

✓

Dataset Build Completed

Build for "dice_L" completed (train=7518, val=940, test=940).

Aug 06, 2026, 04:38 PM

✓

Dataset Scan Completed

Scan for "asb12_260106" completed with status READY.

Aug 06, 2026, 03:54 PM

✓

Dataset Scan Completed

Scan for "asb12_260105" completed with status READY.

Aug 06, 2026, 03:54 PM

✓

Dataset Scan Completed

Scan for "asb12_260103" completed with status READY.

Aug 06, 2026, 03:53 PM

✓

Dataset Scan Completed

Scan for "asb12_260104" completed with status READY.

Aug 06, 2026, 03:53 PM

✓

Dataset Scan Completed

Scan for "asb12_261225" completed with status READY.

Aug 06, 2026, 03:52 PM

Controls: **Mark all read**, per-item **Mark read**, an **Unread only** checkbox, a severity filter and **Reset filters**. The list loads more as you scroll and shows a total in the header.

Three severities: ✓ **SUCCESS** (green), ! **WARNING** (yellow) and × **ERROR** (red). For example, a scan that finishes cleanly raises a ✓ *Dataset Scan Completed*; one that finishes *INVALID* raises *Dataset Scan Found Problems*.

[Home](#)
[Datasets](#)
[Models](#)
[Training](#)
[Benchmarks](#)
[Jobs](#)

Notifications

[Admin](#)

ADMIN
System Admin

Sign Out

Notifications

1 total

Mark all read

☒ Unread only

All severities

Reset filters

✓

Training Completed

Training "dice_L_X" completed; model registered.

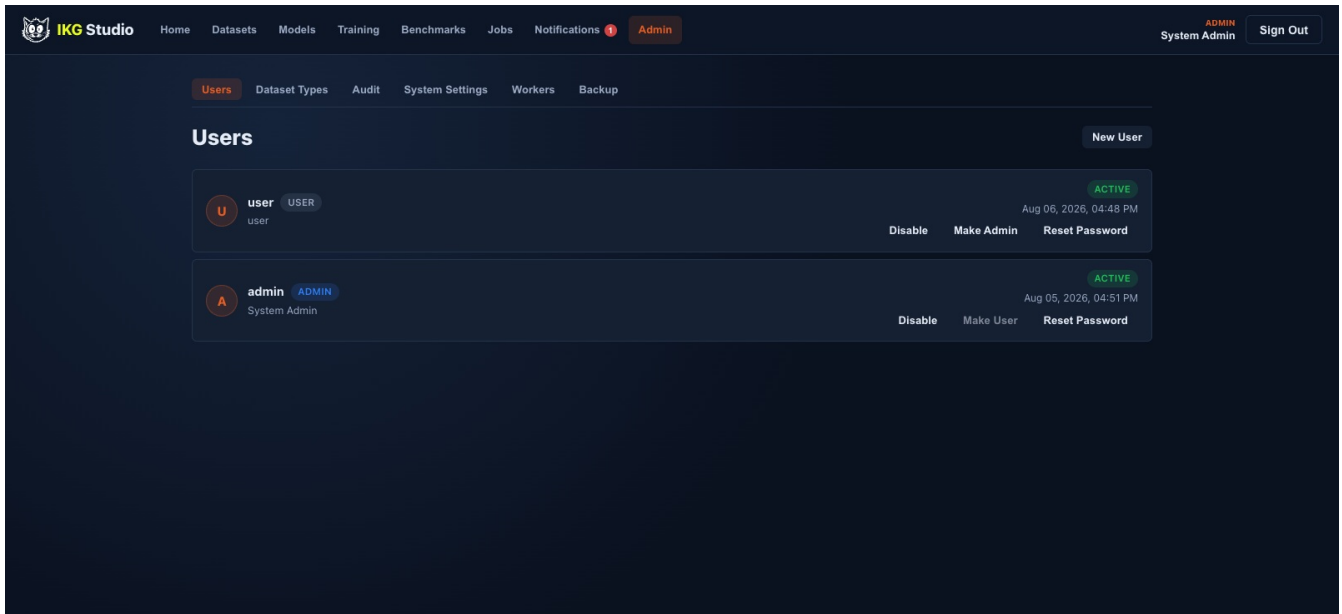
Aug 06, 2026, 08:12 PM

Mark read

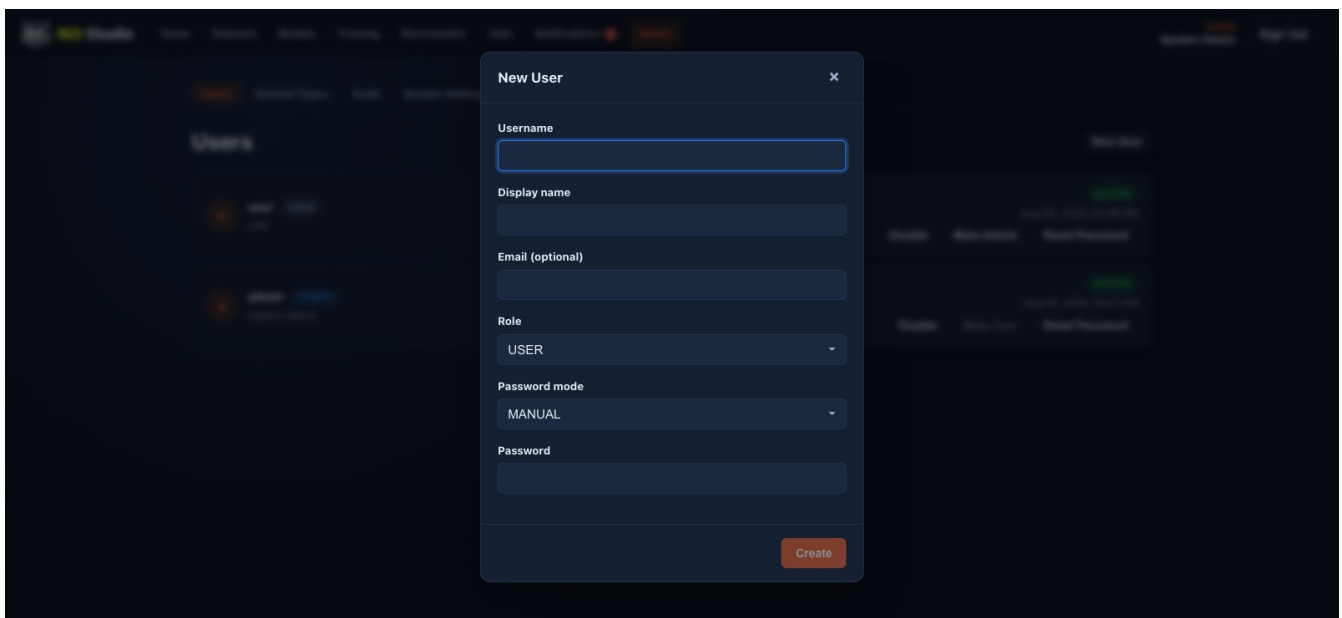
13. Admin

Admins only. Six sub-tabs.

13.1 Users



Accounts have status **ACTIVE**, **DISABLED** or **LOCKED**, and the available actions depend on it: **Disable / Enable**, **Make Admin / Make User** (never on your own account), **Reset Password**, and **Unlock** for a locked account.

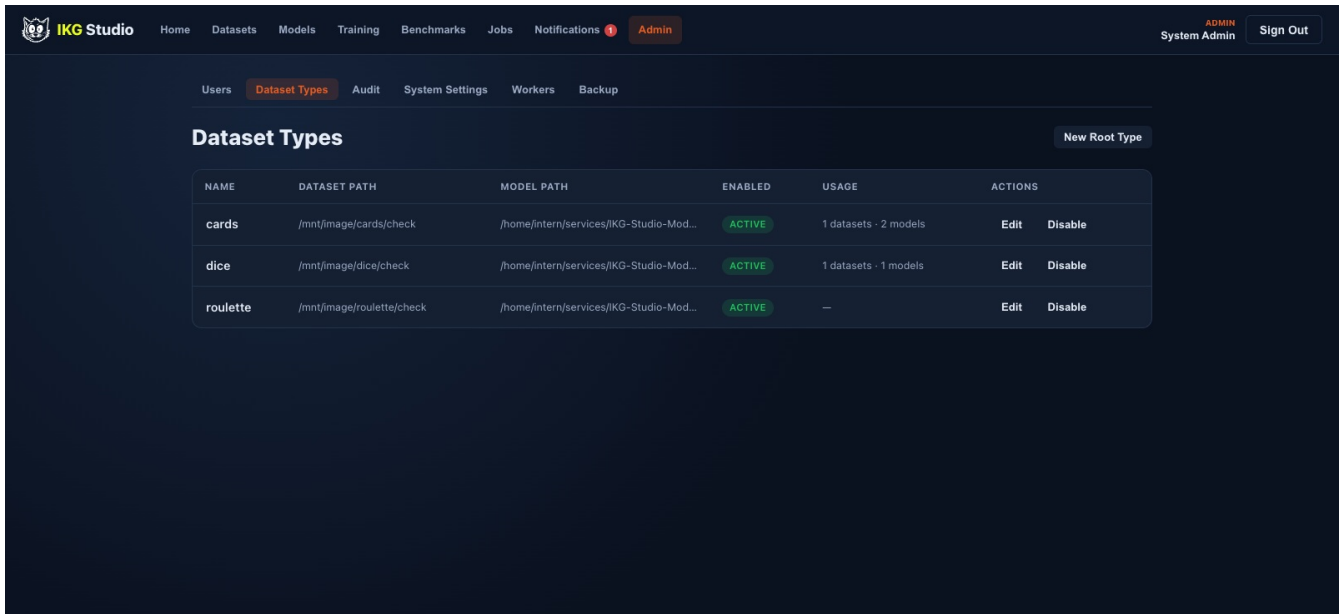


+ **New User** takes a username, display name, optional email, role, and a password mode.

Use "Manual" password mode. In this build the *Generated* mode creates the account but never shows you the generated password, leaving the user unable to sign in. Set a password yourself and hand it over.

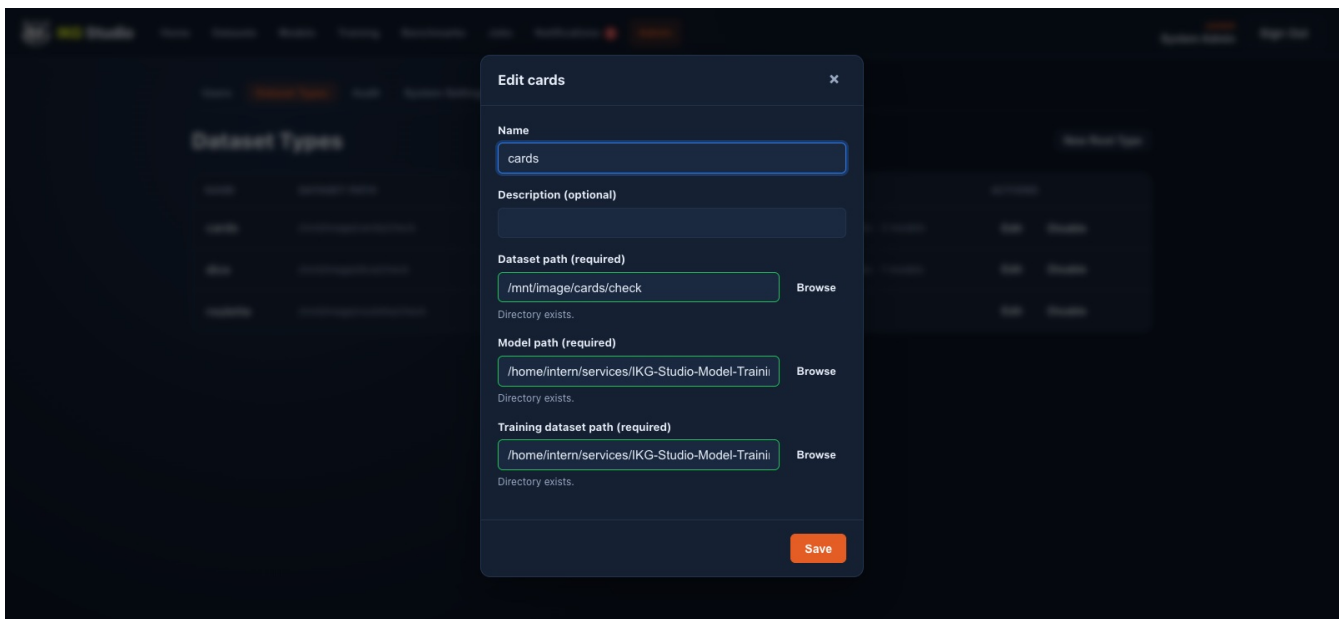
The platform refuses to demote or disable the **last active admin**.

13.2 Dataset Types



Dataset types form a **tree**, not a flat list — rows expand, children are indented, and depth is capped by `dataset_type_max_depth` (default 8).

Each node carries three absolute filesystem roots — **dataset path** (source folders), **model path** (`.pt` files) and **training dataset path** (built or registered YOLO directories) — plus an enabled flag and usage counts. A node without its own path **inherits** the nearest ancestor's, displayed as "*(inherited)*".



Edit changes the name, description and the three paths. Each path is checked for existence as you type, has a **Browse...** picker, and must differ from the other two. Concurrent edits are rejected rather than silently overwritten.

Other actions: **Disable / Enable**, and **Delete** — refused while any child type, source dataset, training dataset or model still references the type, and it names the blockers. Deleting removes only the type and its cached folder index; nothing on disk is touched. Types marked **system** are read-only.

New Root Type creates a top-level type. In this build child types can only be created through the API.

13.3 Audit

 IKG Studio

[Home](#)[Datasets](#)[Models](#)[Training](#)[Benchmarks](#)[Jobs](#)[Notifications](#)[Admin](#)

ADMIN

System Admin

Sign Out

[Users](#)[Dataset Types](#)[Audit](#)[System Settings](#)[Workers](#)[Backup](#)

Audit Log

Action

e.g. user.create

Resource type

e.g. DATASET_TYPE

Result

Any

Actor type

Any

From

yyyy/mm/dd, --:-- --

To

yyyy/mm/dd, --:-- --

No filters applied - 1138 entries

ResetExport CSV

TIME	ACTOR	ACTION	RESOURCE	RESULT	CORRELATION
Aug 07, 2026, 09:33 AM	admin	Login succeeded AUTH_LOGIN_SUCCEEDED	USER f76451ea	SUCCESS	67099db9
Aug 07, 2026, 09:32 AM	login	Login failed AUTH_LOGIN_FAILED	USER f76451ea	FAILURE	a5a5e7a0
Aug 07, 2026, 09:18 AM	admin	Login succeeded AUTH_LOGIN_SUCCEEDED	USER f76451ea	SUCCESS	60f5eea0
Aug 06, 2026, 08:12 PM	training-worker-1	Model created MODEL_CREATED	MODEL 82e82d3c	SUCCESS	77ce8d0d

Every auditable action — logins, resource creation, job lifecycle — with actor, action, resource type and id, result, and a correlation id. Filter by action, resource type, result, actor type and date range, and **Export CSV**. Clicking a **correlation id** opens every event from the same operation across subsystems.

Per-resource **History** cards on models, training jobs and dataset types show the same data scoped to one record, and are visible to USERS too.

13.4 System Settings


[Home](#)
[Datasets](#)
[Models](#)
[Training](#)
[Benchmarks](#)
[Jobs](#)
[Notifications](#)
[Admin](#)

[Admin](#)
[System Admin](#)
[Sign Out](#)

[Users](#)
[Dataset Types](#)
[Audit](#)
[System Settings](#)
[Workers](#)
[Backup](#)

System Settings

Applied platform-wide. Changes take effect on the next operation that reads the setting.

AUTHENTICATION	
Failed login attempts before lockout <code>auth_failed_login_threshold</code>	5
Account lockout duration in minutes <code>auth_lockout_minutes</code>	15
Session absolute timeout in hours <code>auth_session_absolute_hours</code>	24
Session idle timeout in minutes <code>auth_session_idle_minutes</code>	480

MODELS	
Allow http:// (non-TLS) model download URLs <code>model_download_allow_http</code>	☐ Disabled
Allow download URLs resolving to private/loopback networks (admin-controlled, SSRF risk) <code>model_download_allow_private</code>	☐ Disabled
Minimum accepted model file size in bytes <code>model_min_size_bytes</code>	9.5367431640625e-7 GB
Global root path for all model files (best.pt dual-store target) <code>model_root</code>	not set
Maximum accepted browser-uploaded model file size in bytes (default 2 GiB) <code>model_upload_max_size_bytes</code>	2 GB

DATASETS	
Maximum dataset type tree depth <code>dataset_type_max_depth</code>	8
Global root path for built managed datasets (split/data.yaml output) <code>managed_dataset_root</code>	not set

WORKERS & QUEUE	
Queue wait time before warning <code>queue_wait_warning_minutes</code>	30
Worker offline timeout in seconds <code>worker_offline_timeout_seconds</code>	90

STORAGE	
Maximum allowed MinIO storage limit in bytes (0 for unlimited) <code>storage_minio_limit_bytes</code>	100 GB
Storage usage warning threshold <code>storage_warning_threshold_percent</code>	85
Workspace file retention hours <code>workspace_retention_hours</code>	24
Global root path for training/benchmark temporary workspace <code>workspace_root</code>	not set

Grouped by category:

- **Authentication** — `auth_failed_login_threshold` (5), `auth_lockout_minutes` (15), `auth_session_idle_minutes` (480), `auth_session_absolute_hours` (24)
- **Models** — `model_download_allow_http`, `model_download_allow_private` (both SSRF-risk toggles), `model_min_size_bytes`, `model_root`, `model_upload_max_size_bytes` (2 GiB)
- **Datasets** — `dataset_type_max_depth` (8), `managed_dataset_root`
- **Workers & queue** — `worker_offline_timeout_seconds` (90), `queue_wait_warning_minutes` (30)
- **Storage** — `storage_minio_limit_bytes` (100 GiB), `storage_warning_threshold_percent` (85), `workspace_retention_hours` (24), `workspace_root`

Settings are saved **one row at a time**: edit a field and that row grows **Revert** and **Save** buttons, with an "*N* unsaved" badge in the header. Byte values get an MB/GB/TB unit selector. Secret values show as *hidden* and cannot be edited here. Changes take effect on the next operation that reads the setting.

13.5 Workers

Workers

Worker Name	Status	Details	Heartbeat	Registered
dataset-worker-1	ONLINE	DATASET · c6c921686913 · 0 active Python 3.12.12 · torch · ultralytics	Aug 07, 2026, 09:44 AM	Aug 05, 2026, 04:51 PM
training-worker-1	ONLINE	TRAINING · a27f494fa5d6 · 0 active Python 3.12.12 · torch 2.6.0+cu124 · ultralytics 8.4.106 · CUDA 12.4	Aug 07, 2026, 09:43 AM	Aug 05, 2026, 04:51 PM

A read-only list of registered workers: type, online/offline, hostname, active job count, runtime versions (Python, torch, ultralytics, CUDA), and heartbeat and registration times. **There is no enable / disable / drain action** — a worker goes offline by stopping, or by missing heartbeats for `worker_offline_timeout_seconds`.

GPU inventory is not shown here; it appears in the device picker when you launch a training job or benchmark.

13.6 Backup

Backup & Restore

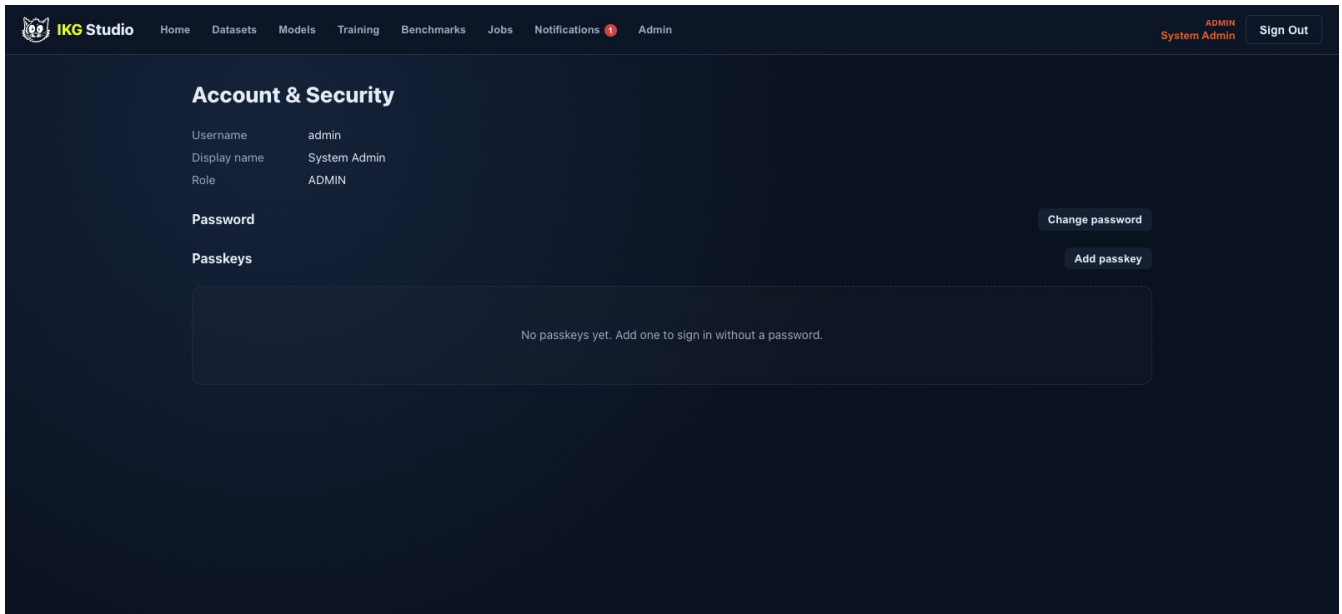
Export downloads a JSON file containing all users (incl. password hashes & passkeys), dataset types and system settings (incl. secret values). Import restores the file by overwriting these tables; it is only supported on a fresh/empty system.

[Export backup](#) [Import backup](#)

Export downloads `ikg-backup-<timestamp>.json` containing all users (including password hashes), dataset types, system settings (including secret values) and registered passkeys. Treat the file as a credential.

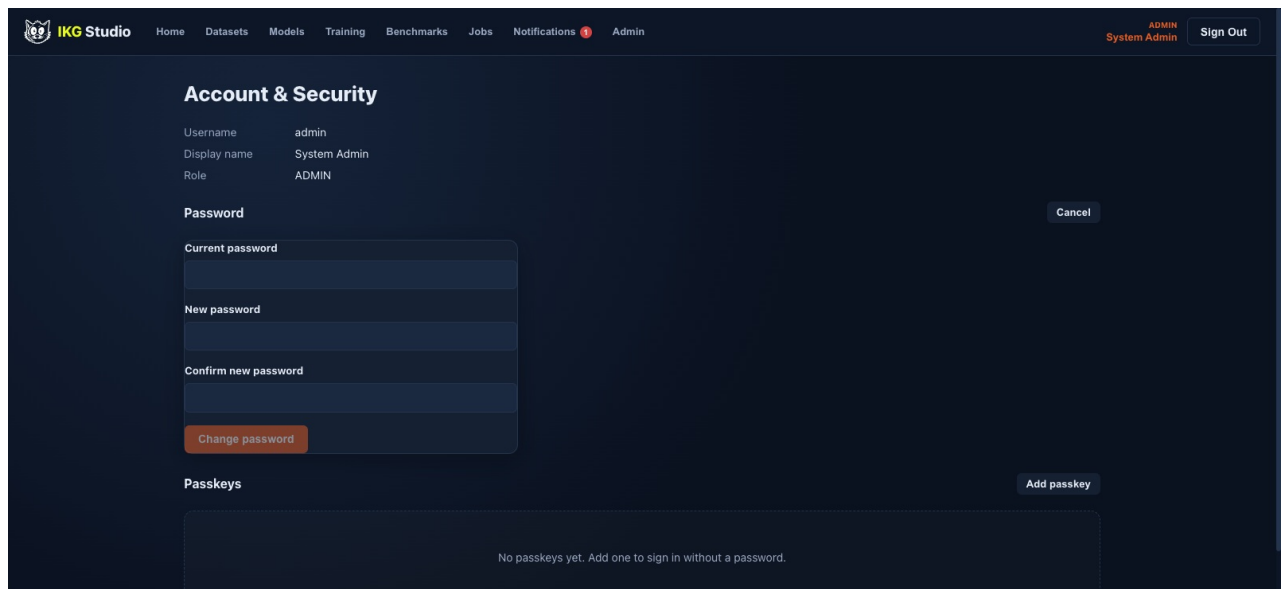
Import replaces those four tables in a single transaction. Your own account survives — it is re-pointed at the matching admin in the file, so your session stays valid — but **every other user is signed out**. The import rolls back if any dataset type is still referenced by a source dataset, training dataset or model, which in practice means it is only usable on a system that has no dataset or model data yet. Datasets, models, jobs, benchmarks, artifacts and audit logs are never touched.

14. Account & Security



Shows your username, display name and role (read-only), plus:

- **Change password** — current password, new password, confirmation.



- **Passkeys** — register WebAuthn credentials for passwordless sign-in, and remove ones you no longer use. A registered passkey is what makes the passkey button on the login page work.

15. Status Reference

Execution status — the technical state of one attempt

ASSIGNED → CLAIMED → PREPARING → RUNNING → SUCCEEDED / FAILED / CANCELLED / STOPPED / LOST

The Jobs page quick filters map onto these: **Active** = assigned, claimed, preparing, running · **Completed** = succeeded · **Failed** = failed, cancelled, stopped, lost.

Business status — the state of the thing being worked on

Job type	Business status comes from
Training	the training job
Dataset Build / Training Dataset Scan	the training dataset
Dataset Scan	the scan , not the source dataset
Benchmark Eval	the evaluation

Model Ingest	the ingest task
Model Conversion	the conversion

Training job

SCHEDULED · QUEUED · PREPARING · RUNNING · STOPPING · COMPLETED · FAILED · CANCELLED · STOPPED · BLOCKED

BLOCKED means the job is waiting for another job to finish. The scheduler promotes it once its dependencies complete, and fails it if one of them fails. Dependencies can only be set through the API in this build, so **BLOCKED** will not appear for jobs created in the UI.

Benchmark run

QUEUED · RUNNING · COMPLETED · PARTIALLY_FAILED · FAILED · CANCELLED · STOPPING · STOPPED

PARTIALLY_FAILED means some evaluations succeeded and some did not; the successful metrics are kept. Note that the list page's status filter does not offer **STOPPING** or **STOPPED**, so a stopped run cannot be filtered for.

Benchmark evaluation

PENDING · QUEUED · RUNNING · COMPLETED · FAILED · CANCELLED · STOPPING · STOPPED

Model

AVAILABLE · DELETED (soft delete — the record stays, the file is unlinked)

User

ACTIVE · DISABLED · LOCKED

16. Troubleshooting & Edge Cases

A job says LOST. The worker stopped sending heartbeats. The scheduler marks the execution **LOST** and then carries the failure into the object being worked on — it does not leave it hanging:

Job type	What happens next
Training	retried automatically with a new attempt, up to a configured limit, then FAILED . If it was already stopping, it becomes STOPPED .
Dataset Scan	the scan fails and the source dataset goes back to REGISTERED — not INVALID , since the scan never actually ran
Dataset Build / Training Dataset Scan	the training dataset becomes INVALID
Benchmark Eval	the evaluation fails and the parent run is finalised so it cannot hang in RUNNING
Model Conversion	the conversion becomes FAILED

The log will show `Execution lost: no heartbeat within timeout`.

A training job shows "attempt 3". Some failures are retried automatically with a growing delay — temporary storage failures, database connection failures, network and Redis timeouts. Everything else (a bad dataset, missing labels, an unreadable model, a checksum mismatch, invalid configuration) is treated as permanent and fails on the first attempt. The **Executions** table on the job detail page shows each attempt and its error.

A detail page says the record no longer exists. Opening a URL with an id that has been deleted — or that you cannot see — shows an explicit error panel with a **← Back** button. There is no 404 page because there are no page URLs; a `?modelId=...` parameter only takes effect if the Models tab is already open.

A source dataset scan finished INVALID. Open the dataset's detail page: the red banner groups the errors by code with an example file and line. Common causes are class indices in the labels that are missing from the class list, and label files with no matching image. Fix the folder, or set a class list override, then **Rescan**.

A source dataset card says the folder is not on disk. Its folder left the directory index. **Rescan** recovers it if the folder is back; otherwise **Archive** it.

"Storage Limit Exceeded" is showing. MinIO is at its quota and new uploads and executions are blocked. Delete unused OpenVINO conversions, models or benchmark artifacts, or raise `storage_minio_limit_bytes`.

A benchmark run finished PARTIALLY_FAILED and I want the failed cells re-run. There is no way to re-run a single cell. Retry re-runs the whole run and clears the metrics of the cells that succeeded before recomputing them — so either accept that, or create a new run for the failing pair.

A new user cannot sign in. If the account was created with the *Generated* password mode, no one ever saw the password. Reset it from Admin → Users with a password you choose.

Repeated failed sign-ins lock the account. Status becomes `LOCKED`; wait `auth_lockout_minutes` or ask an admin to **Unlock**.

17. End-to-End Workflow

1. **Admin** creates a **dataset type** with its three filesystem roots (Admin → Dataset Types).
2. Someone drops raw image folders under the type's **dataset path**.
3. On **Datasets → Source**, expand the type and **Scan & register** the folders. Fix anything that comes back `INVALID`, using a class list override where the folder's `classes.txt` is wrong.
4. On **Datasets → Training**, build a **training dataset** from the source datasets (or register a directory that is already split).
5. On **Training**, launch a **training job**: pick the type, an official or registered model, the dataset, the hyperparameters and the device.
6. Watch it on **Jobs**; the result appears under **Models**.
7. **Download the weights** from the model's Artifacts card — see [Section 8](#) — or convert to OpenVINO if you need IR.
8. On **Benchmarks**, evaluate the model against a dataset, then use **Compare Models** to put candidates side by side and export the comparison.